Direct and indirect winding processes

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Vorlesung Produktion elektrischer Motoren und Maschinen
Lecture Production of electric motors and machines

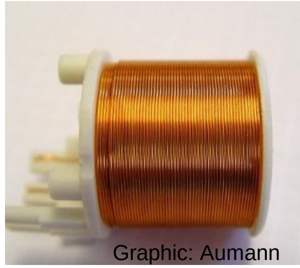
The lecture unit direct and indirect winding processes provides an overview of the fundamental winding variants and their manufacturing processes.

As a primary component of electrical machines, windings have a meaningful influence on the power, efficiency, and space requirements of electric devices.

After attending the lecture “Direct and indirect winding processes” students shall:

- **understand winding packing density and fill factor and differentiate between these values,**
- **describe the technical terminology and process depictions for winding methods,**
- **identify and explain using technical terminology alternating current windings based on their characteristics,**
- **differentiate the manufacturing components and process variants used to produce different types of windings, and**
- **select a suitable winding process for typical electric motor designs.**


Graphic: Essex

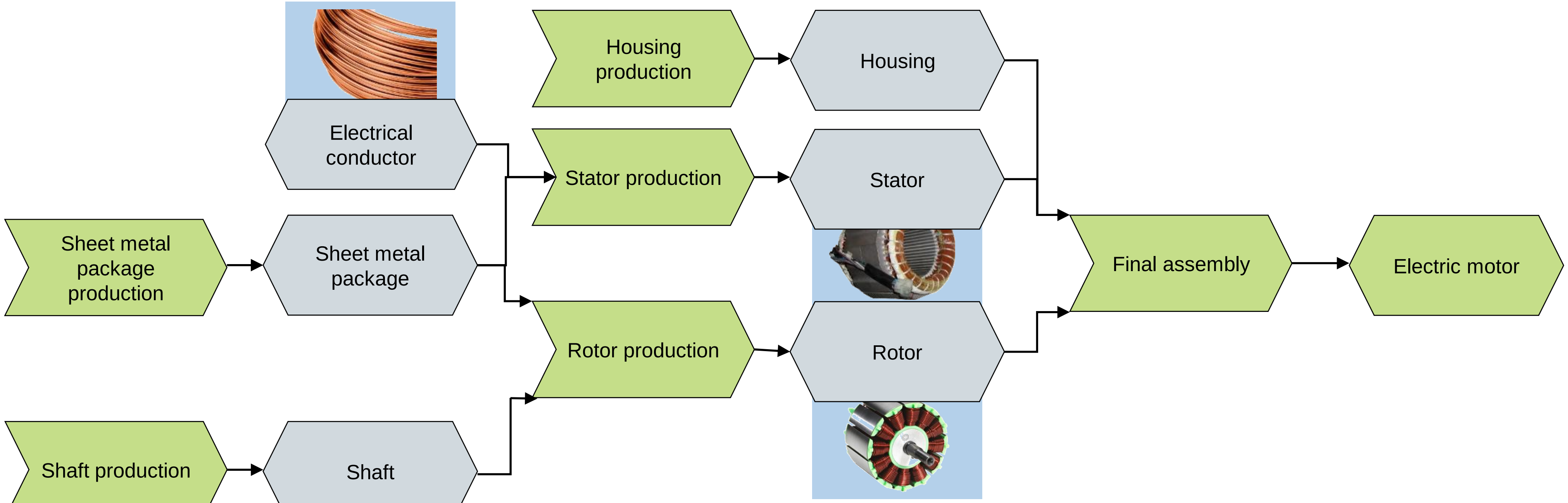

Graphic: Aumann


Graphic: BMW


Graphic: Risomat

Through winding processes, electrical conductors are formed into coils and mounted on the stator or rotor lamellar stack.

Slide from Lec. 1



Graphics: Vacuumschmelze, Aumann, Tarun, Essex

Lecture unit 6 provides an overview of the design and production steps for direct and indirect winding processes.

- **Technical terminology**
  - **Definition of winding components and schematic diagrams**
  - Significance of packing density and fill factor
  - Important characteristics of multi-phase alternating current windings

- Winding processes and corresponding production system technology
  - Components of a winding system
  - Production diagrams for different winding processes
  - Integration of the most common winding processes into the production chain for stators

Graphic: ZF Friedrichshafen AG

A complete motor winding is composed of individual wires in turns, coils, coil groups, and phases.

Winding

Made from several phases
(3 depicted)

Phase	Phase	Phase

Typically made from several spool groups
(1 depicted)

Coil group

Made from several coils

(2 depicted)

Coil	Coil

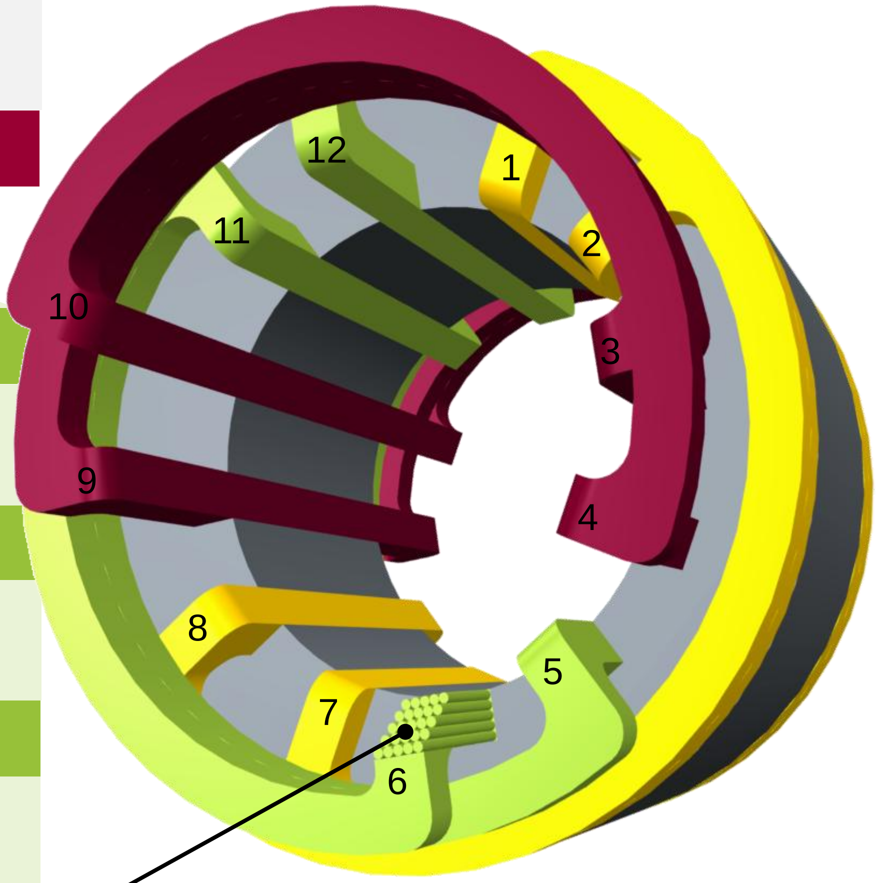
Made from many turns of wire

(25 depicted)

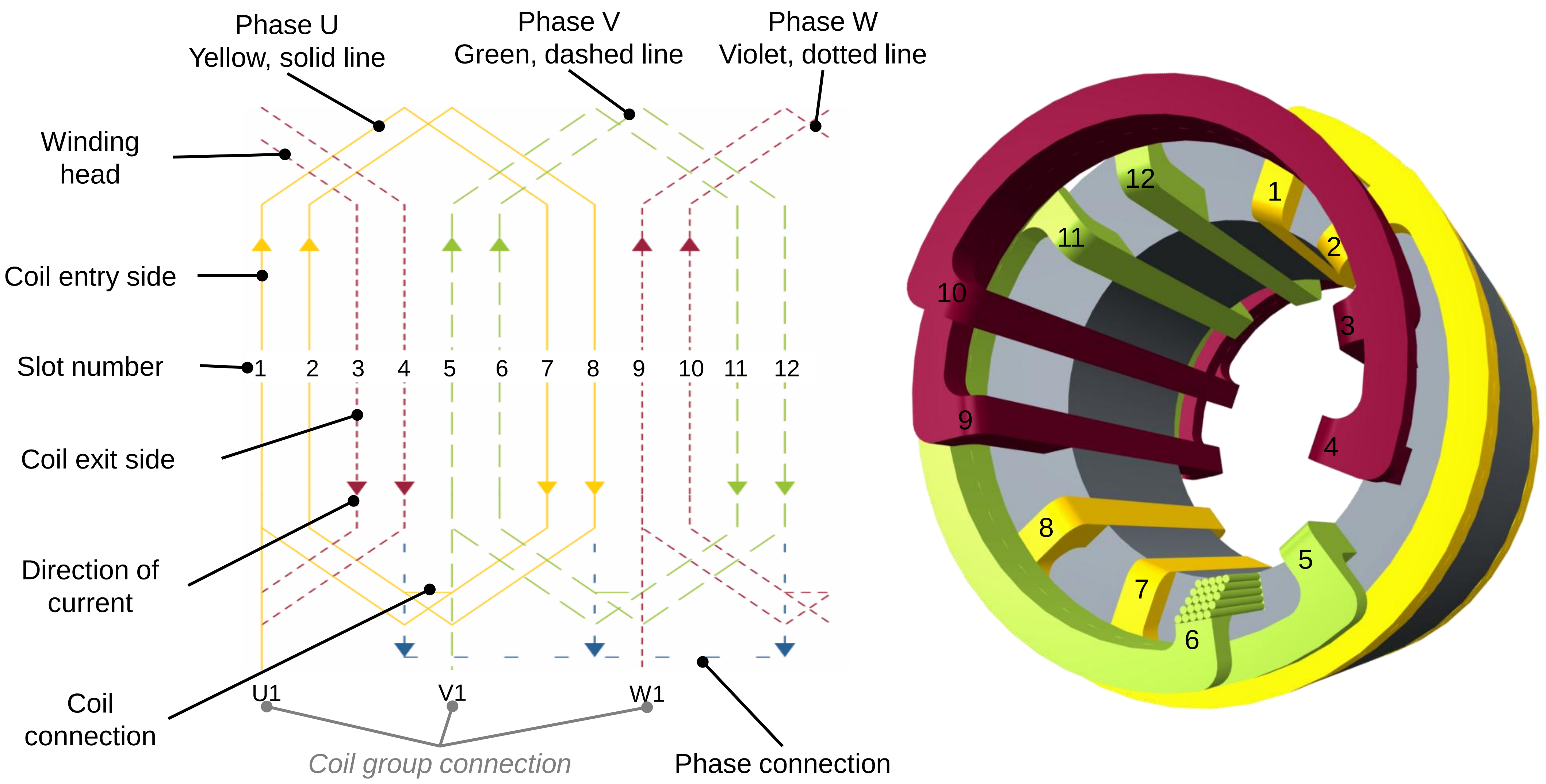
Turn	Turn	Turn	Turn

made from a ‘forward’ and ‘reverse’ current carrying length of wire

Wire	Wire	Wire	Wire	Wire	Wire	Wire	Wire



Windings in rotating electrical machines are depicted using an unrolled lateral surface view to make the winding configuration easier to identify and for documentation.



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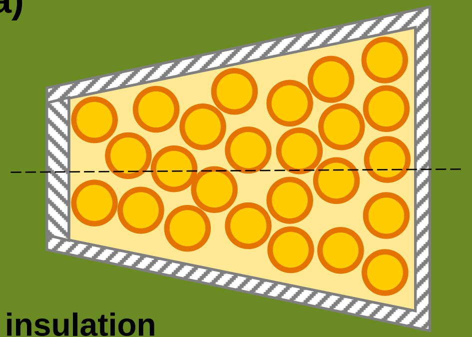
Graphic: ZF Friedrichshafen AG

Fill factor of an electrical winding may be calculated using different methods and attention must be paid to the definition of the fill factor in use.

Slide from Lec. 4

Slot fill factor (square wire area)

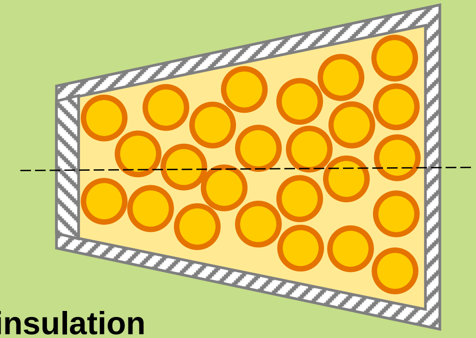
$$k_N = \frac{z_n \cdot \mathbf{d_i^2}}{A_{n\,i}}$$



with
- z_n – Number of wires per slot
- d_i – Outer diameter of **wire and insulation**
- $A_{n\,i}$ – Cross section of the **insulated slot**

Mechanical fill factor

$$k_{mech} = \frac{\mathbf{A_{Wire,i}}}{A_{n\,i}} = \frac{\frac{1}{4} \cdot d_i^2 \cdot \pi \cdot z_n}{A_{n\,i}}$$

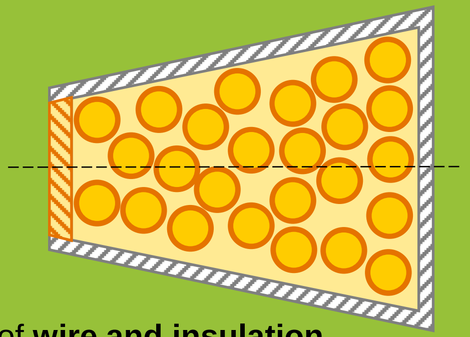


with
- z_n – Number of wires per slot
- d_i – Outer diameter of **wire and insulation**
- $A_{n\,i}$ – Cross section of the **insulated slot**

Fill factor

Winding fill factor

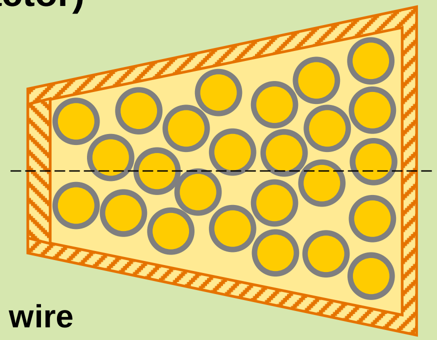
$$k_W = \frac{z_n \cdot d_{i\,max}^2}{A_{n\,eff}}$$



with
- z_n – Number of wires per slot
- $d_{i\,maxx}$ – **Largest** outer diameter of **wire and insulation**
- $A_{n\,eff}$ – Cross section of the insulated slot **and slot opening**

Electrical fill factor (copper fill factor)

$$k_{elec} = \frac{\frac{1}{4} \cdot d_{Cu}^2 \cdot \pi \cdot z_n}{A_n}$$



winding
- z_n – Number of wires per slot
- d_{Cu} – Nominal diameter of the **bare wire**
- A_n – **Total** slot cross section

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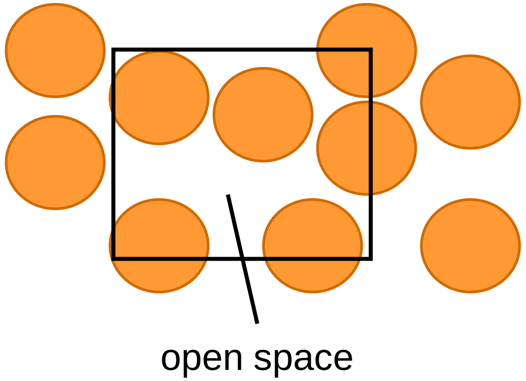
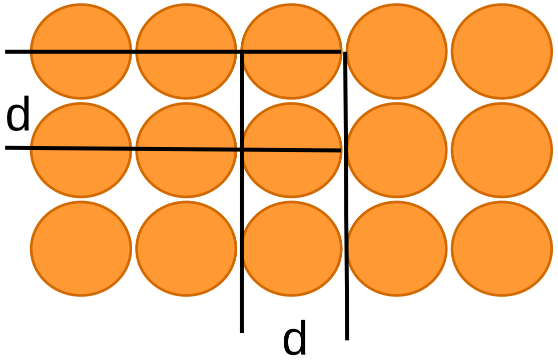
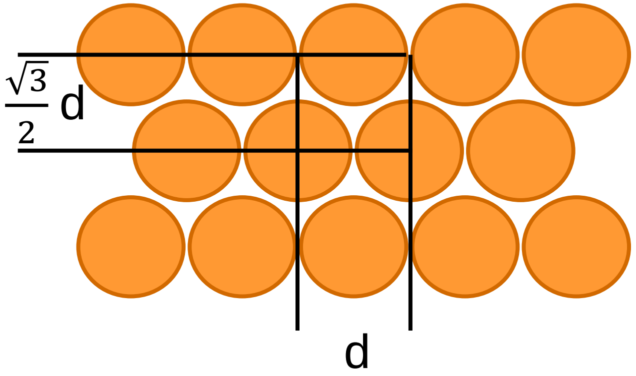
The packing density as a ratio of the slot cross-section to the total cross-section of the conductors reflects the quality of the layer structure within a winding, not within a slot!

Wild winding	Linear winding	Orthocyclic winding



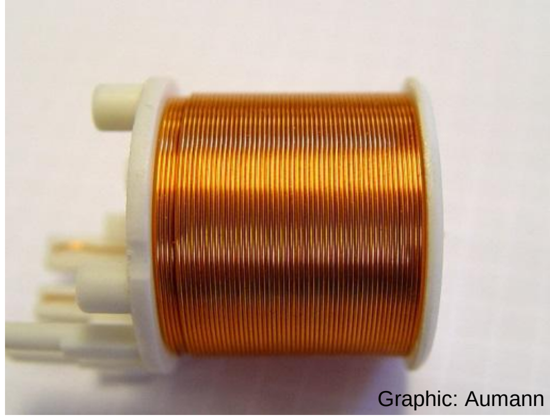




Max. Packing density k_{mech}

Wild winding	Linear winding	Orthocyclic winding
65 – 75%	78.5%	90.7%

Precise winding arrangement as well as higher fill factor greatly influence the overall motor characteristics.



Graphic: Aumann

Precise winding arrangement

High fill factor

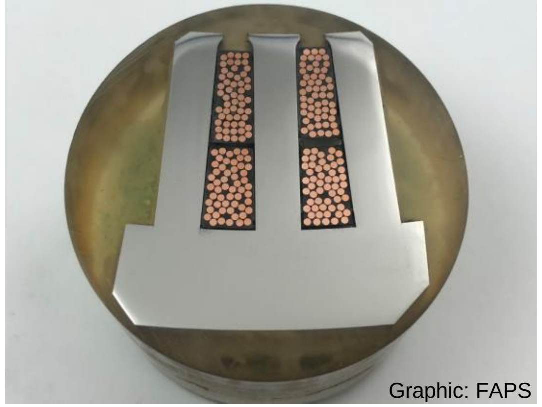
- **Improved efficiency**
- **Higher power density**
  - Reduced size
  - Higher power

Improved thermal and mechanical properties

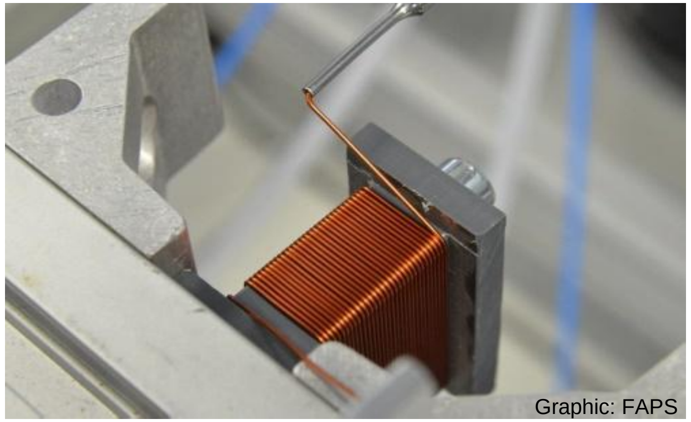
- Mechanical stability
- Thermal conductivity

Improved electric and electromagnetic properties

- Less frequent electrical breakdown
- Constant wire length



Graphic: FAPS



Graphic: FAPS

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Graphic: ZF Friedrichshafen AG

Windings may be modified and categorized according to a variety of geometric design factors used in the coil shape and arrangement.

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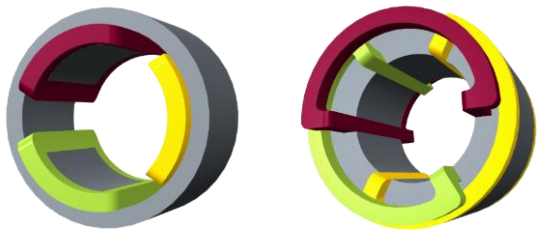
Winding classification according to geometric design factors

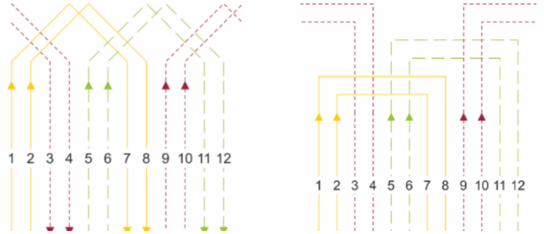
Distribution of windings along the perimeter

- Concentric windings
- Distributed windings



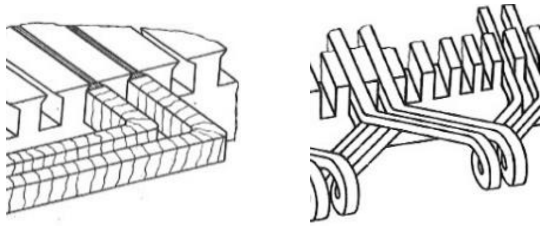
Coil width of coils in a coil group

- Equal width
- Unequal width



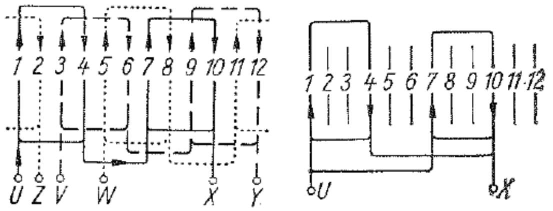
Number of coil side layers

- Single-layer winding
- Double-layer winding



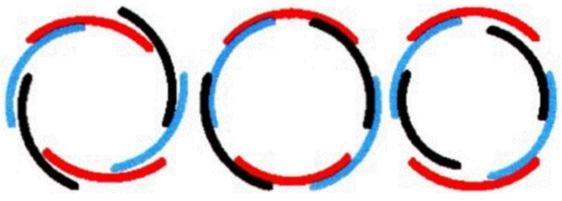
Coil group connection

- Series connected
- Parallel connected



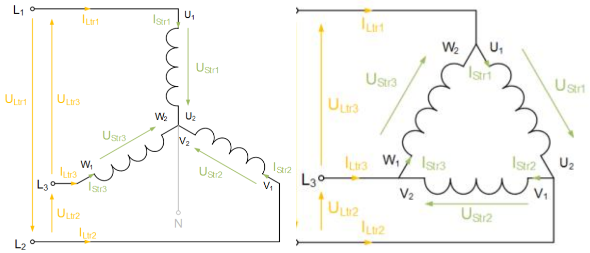
Coil head shape

- One tier
- Two tiers
- Three tiers



Phase connection

- Wye (or star) connection
- Delta connection



Graphics: Bălă, eigene Darstellung

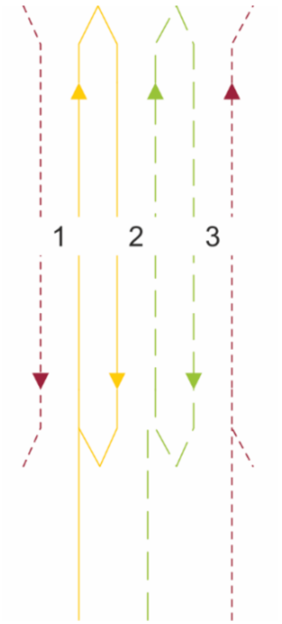
Windings for three-phase machines may use one of two types of designs.

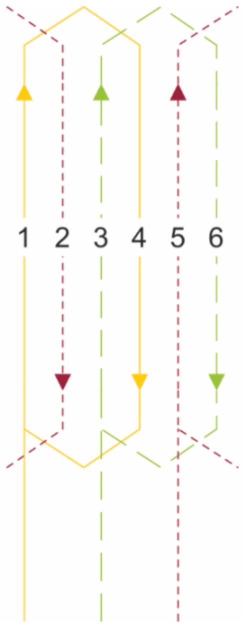
Concentric windings

- Coils do not overlap in the winding head
- Single-layer arrangement:
  one coil side per slot
- Double-layer arrangement:
  two coil sides per slot

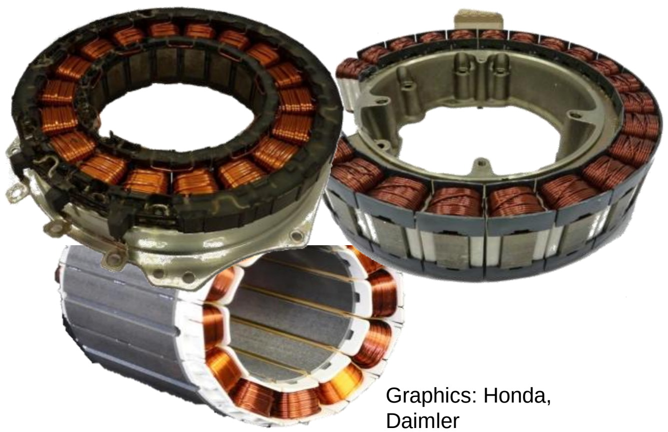
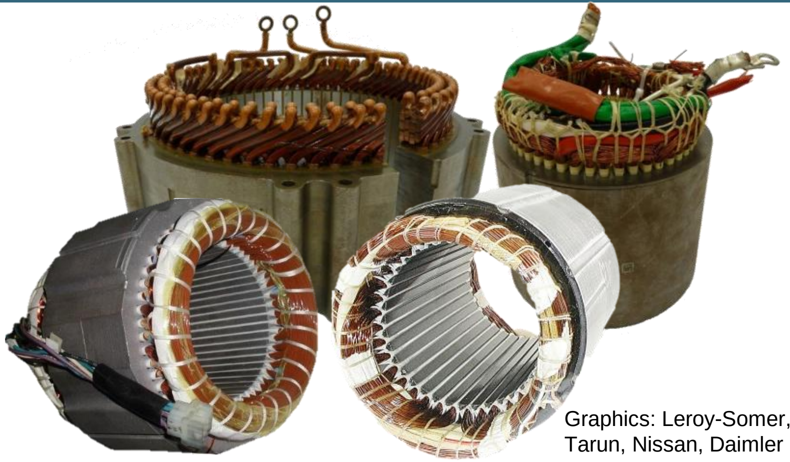
Distributed windings

- Coils overlap each other, distributed over the stator circumference
- Coils “skip” several slots
- Single- and double-layer arrangements possible





Distributed windings are more advantageous during motor operation, however concentric windings are significantly easier to manufacture.

Concentric windings	Distributed windings
 Graphics: Honda, Daimler	 Graphics: Leroy-Somer, Tarun, Nissan, Daimler

Advantages

Concentric windings	Distributed windings
Short winding head	Sinusoidal air gap field (fewer harmonics)
Low production cost	Usable in most machine types
Typically have higher copper fill factor	

Disadvantages

Concentric windings	Distributed windings
Many harmonics in the air gap field, torque ripple, cosφ ↓	Large winding head
Higher rotor losses in asynchronous machines (non-uniform MMF-distribution)	Difficult to manufacture

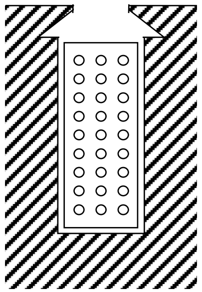
Two-layer winding has design advantages, such as lower noise emissions, which are particularly effective with large machines.

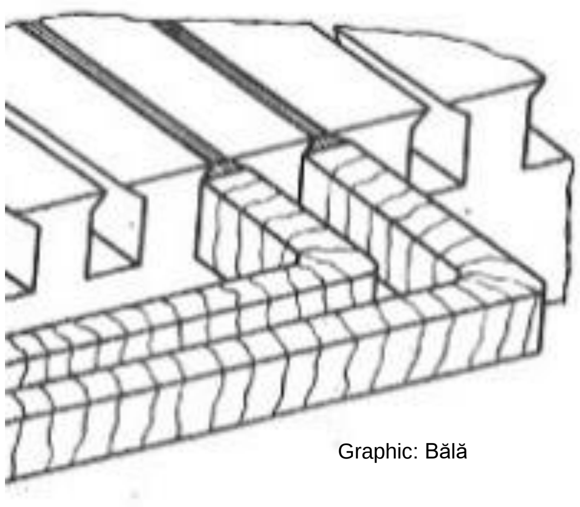
Single-layer winding

- Only one coil side per slot
- Number of individual coils is equal to half the number of stator slots
- Most common stator winding

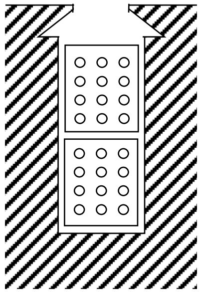
Double-layer winding

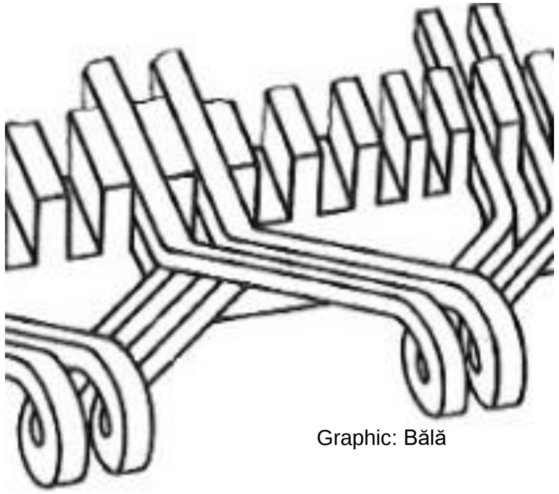
- Two coil sides from different coils stacked into each slot
- Total number of coils is equal to the number of stator slots
- Upper and lower layers separated by insulating layer
- Reduces harmonics





Graphic: Bălă





Graphic: Bălă

Three different coil arrangements are available to form the winding head.

One-level winding head

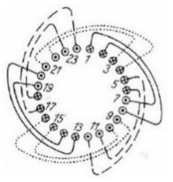
- Wreath, crown, basket, spiral or involute windings
- Especially suited for manual insertion
- Perfect phase symmetry

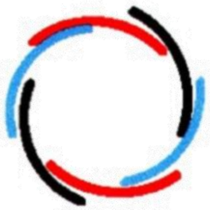
Two-level winding head

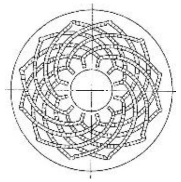
- Insertion in two steps: outer, inner coils
- Compensation for different coil lengths possible
- Most common arrangement

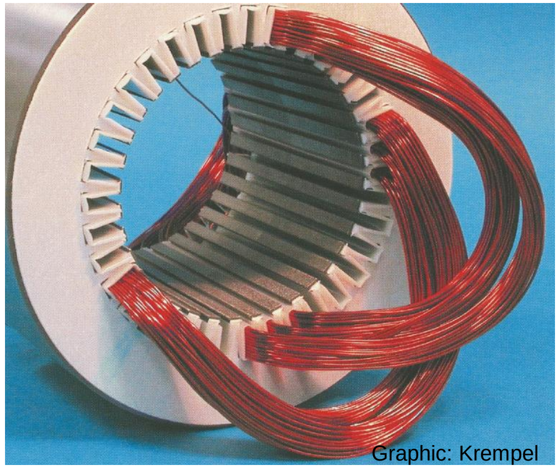
Three-level winding head

- Insertion in three steps: outer, middle, inner coils
- Inserting the series connection between coil groups possible

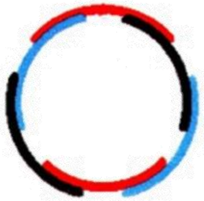


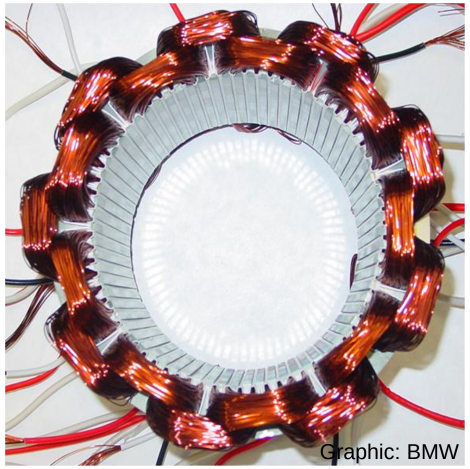




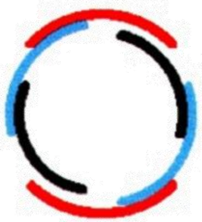


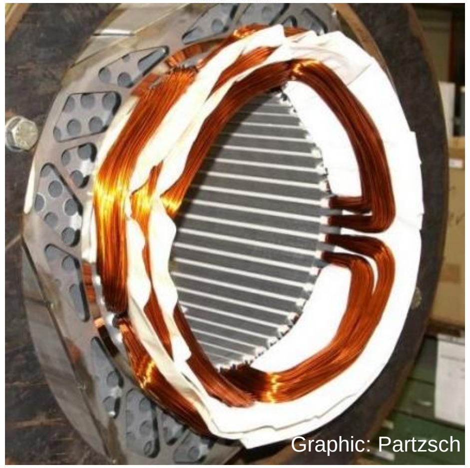
Graphic: Krempel





Graphic: BMW

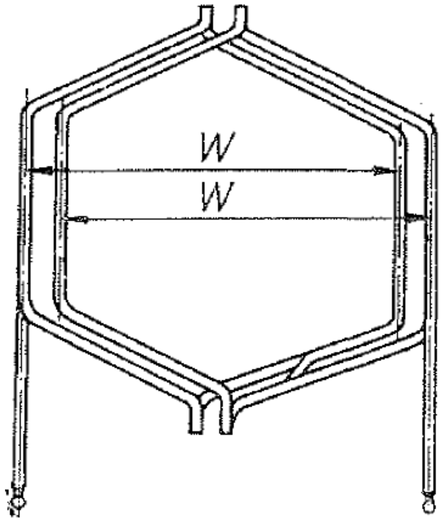




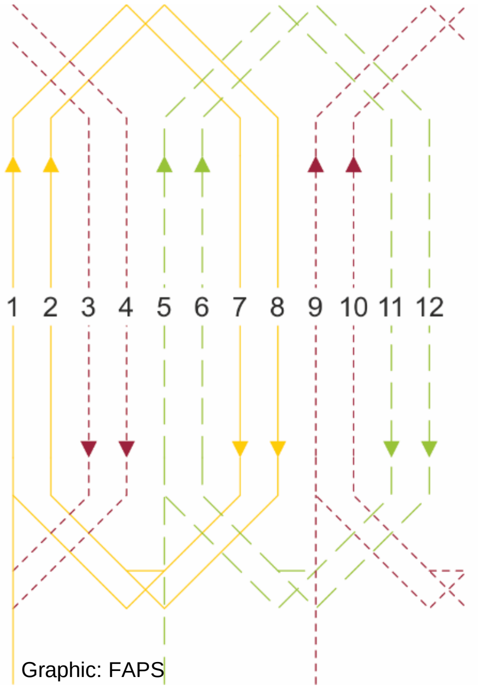
Graphic: Partzsch

A coil group can be arranged with coils of equal width or with concentric coil groups.

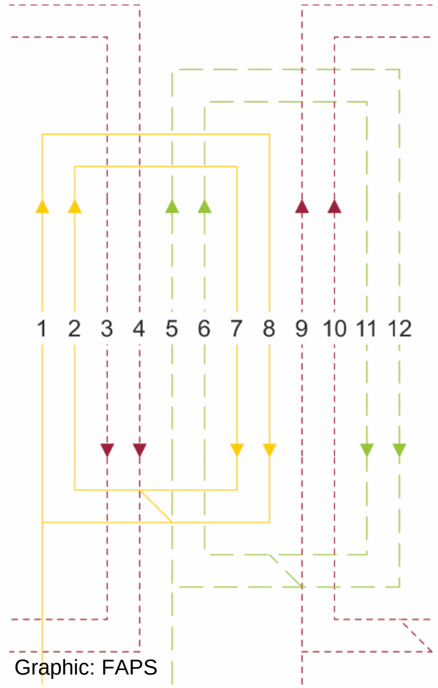
Equal width	Unequal width
■ Coils in a coil group have equal lengths ■ Crossovers occur in the winding head ■ Typical winding head: One level (basket winding)	■ Coils create concentric coil groups ■ Coils within a coil group have unequal lengths ■ Typical winding head: two or three level



Graphic: Bălă



Graphic: FAPS



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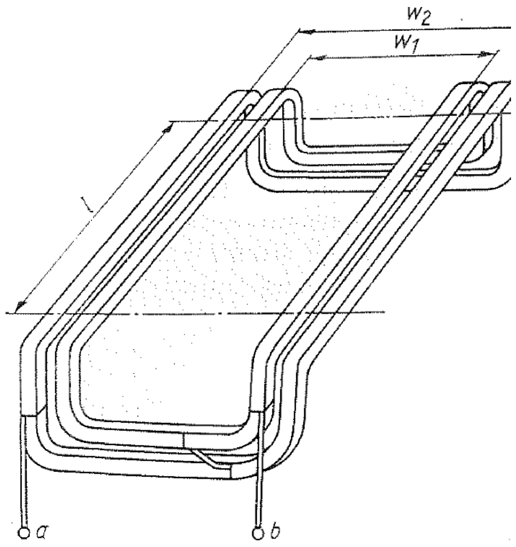


Graphic: Bălă

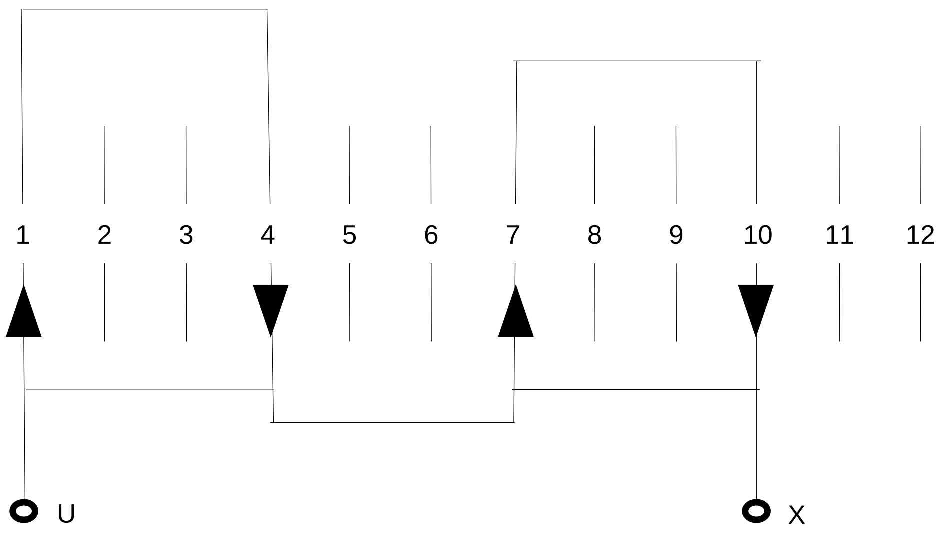
Coil groups are connected in series or parallel according to the desired phase current and voltage.

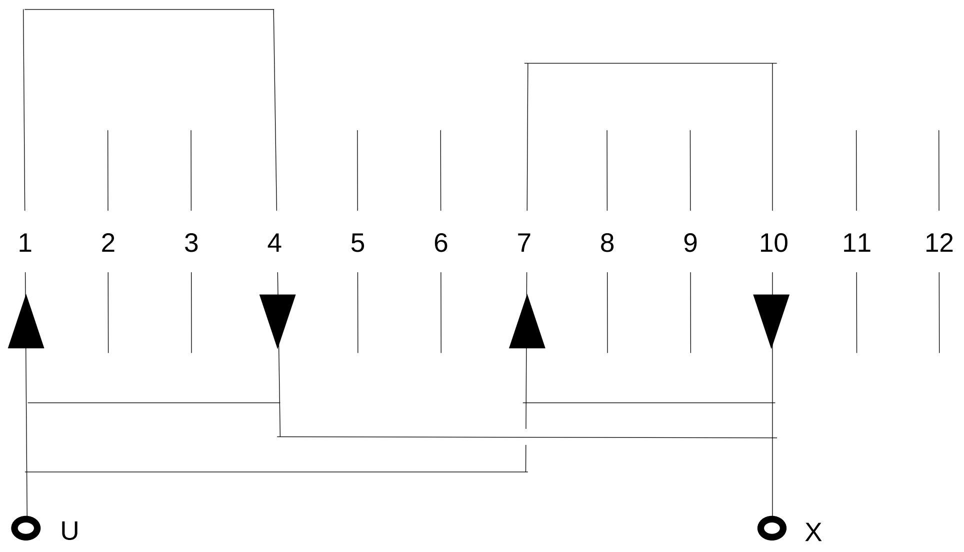
Series connection

- Phase voltage determined by the total number of turns in all series connected coils
- Continuous-winding of coil groups in series possible

Parallel connection

- Phase current distributed through the parallel connection of coils within a phase (current divider)
- Smaller wire dimensions possible





The phases of an electric machine supplied with alternating current may be operated with a star or delta connection.

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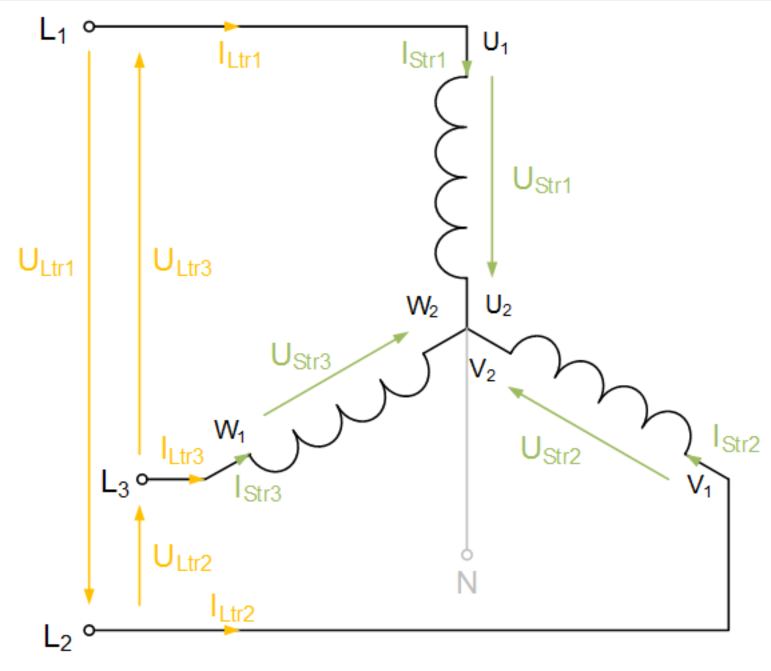
Wye/Star connection

Phase voltage is lower than line voltage by a factor of √3



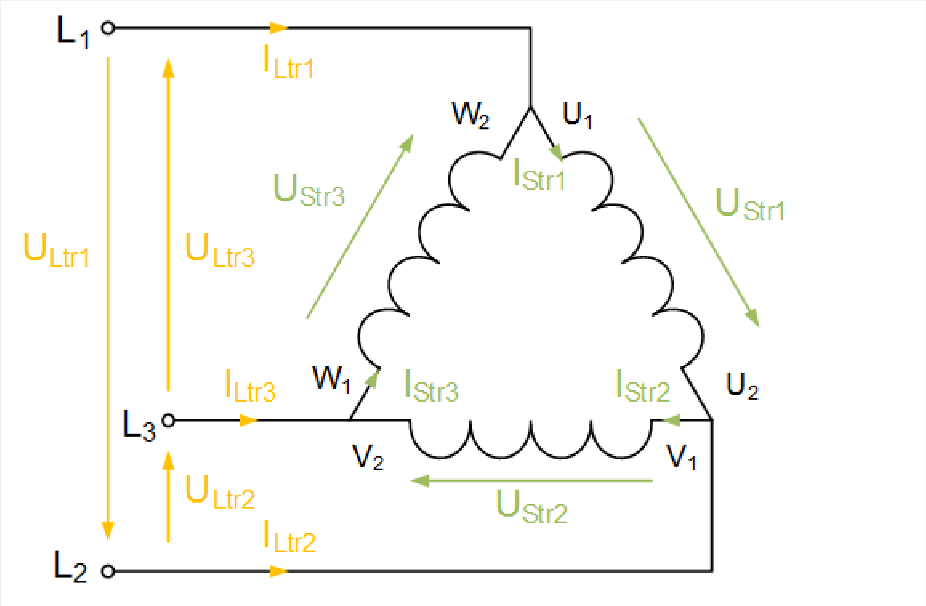
$$U_{Ltr(x)} = \sqrt{3}\, U_{Str(x)} \text{ [V]}$$

$$I_{Ltr(x)} = I_{Str(x)} \text{ [A]}$$

$$P = 3 \cdot U_{Str}\, I_{Str} = \sqrt{3} \cdot U_{Ltr} I_{Ltr}\ [W]$$

Delta connection

The connected network voltage (line voltage) is supplied directly to the phases



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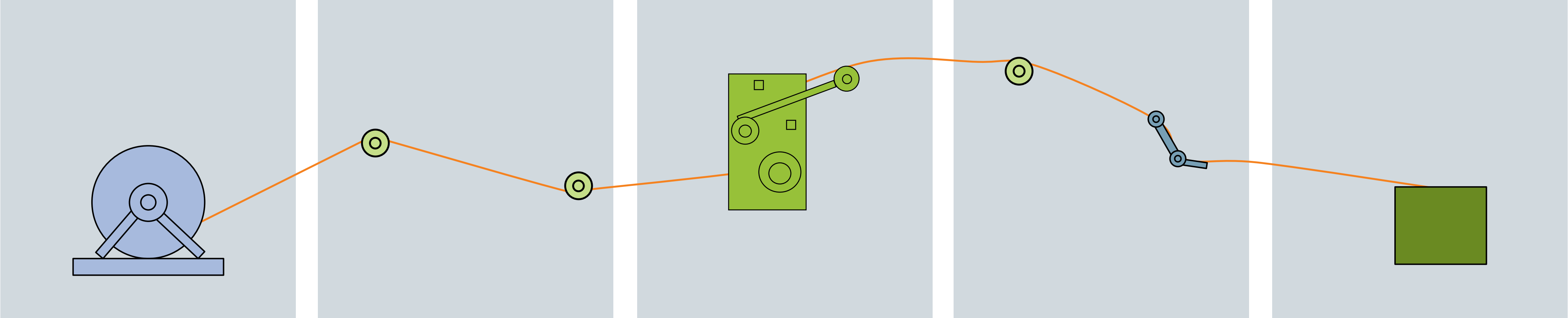
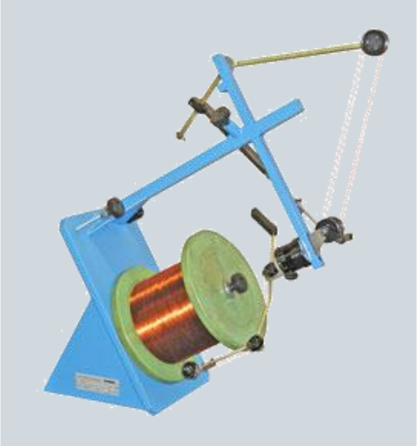
Graphic: ZF Friedrichshafen AG

The different winding methods all possess the same structure of the winding process.

Supply from storage reel	Redirection and guide elements	Wire brakes and compensating elements	Wire guides	Winding body





Graphic: Mobac

Graphic: Mobac

Graphic: LJU

Graphic: Dongguan

Graphic: PVS

The different winding methods all possess the same structure of the winding process.

Supply from storage reel

Redirection and guide elements

Wire brakes and compensating elements

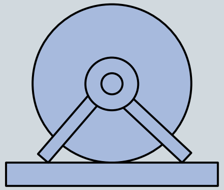
Wire guides

Winding body

Graphic: Heinrich Schümann GmbH & Co. KG

The type of spools and feed devices used to supply wire are strongly determined by the shape of the wire material.

Supply from storage reel

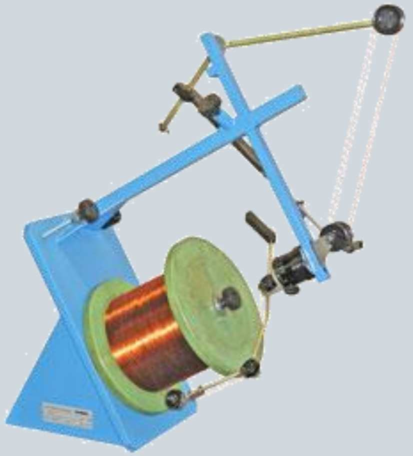


Graphics: hnmagnetwire, astroplast

Flyer operation

Suitable for:

Round wire

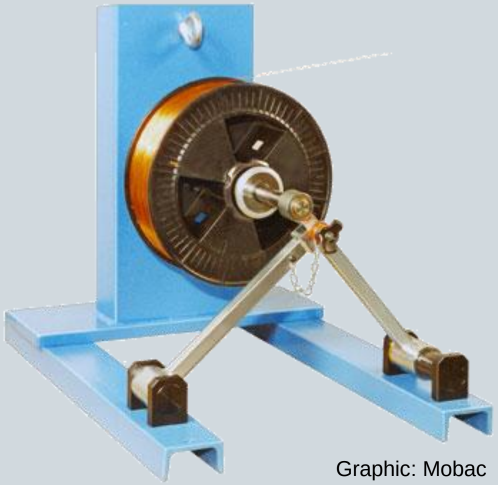


Graphic: Mobac

Tangential operation

Suitable for:

- Round wire
- Flat wire



Graphic: Mobac

Axial operation

Suitable for:

Round wire

Graphic: Mobac

The wire must be deflected and guided in a defined manner several times on its way from the barrel or spool.

Redirection and guide elements

Guiding pulleys

- Redirecting at direction-changing points
- Direction changes possible in one spatial direction per pulley
- No application of force

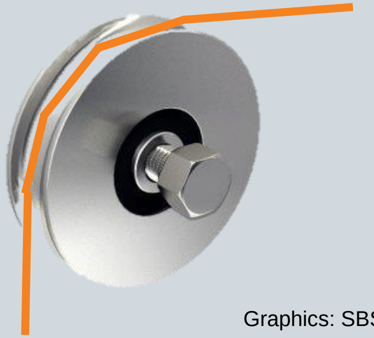
Guiding elements/rails

- Guide the wire flat
- Very smooth surface required
- Lower roughness and possibly hardness than enamel insulation of the wire

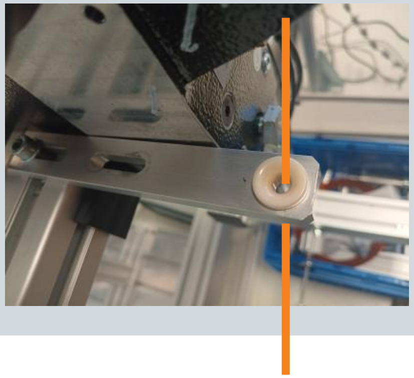
Matrices

- Selective guidance of wire
- Does not prevent twisting or tilting
- Tangential inlet possible
- Usually found in front of needles or guiding pulleys

Graphic: Mobac

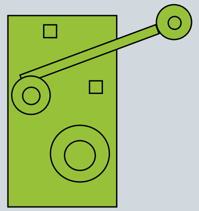


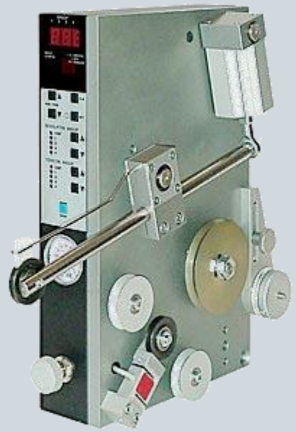
Graphics: SBS



Different actuator principles are used
for wire brakes and compensating elements.

Wire brakes and compensating elements

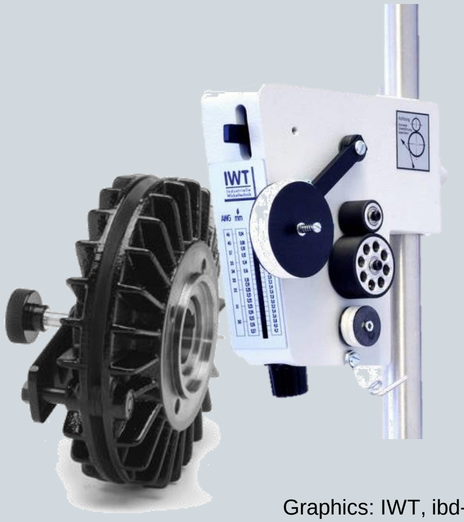




Graphic: LJU

Mechanical

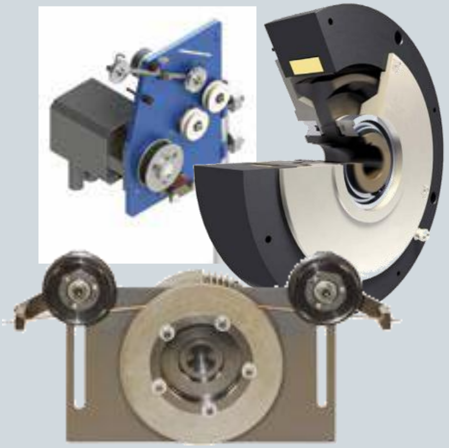
- Spring system
- Sliding block system (disc brakes)



Graphics: IWT, ibd-wt

Electromechanical

- Magnetic particle brakes
- Hysteresis brakes/ eddy current brakes
- Servomotors



Graphics: Mobac, ibd-wt, RUFF

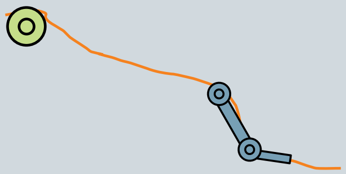
Pneumatic/Hydraulic

- Pneumatic cylinder
- Hydraulic cylinder

Graphics: Re, goHydraulik

The wire guide has the task of directing the wire with as little friction as possible and thus shortening free wire runs.

Wire guides



Graphic: Dongguan

Rollers

Grooved rollers

Graphics: FAPS

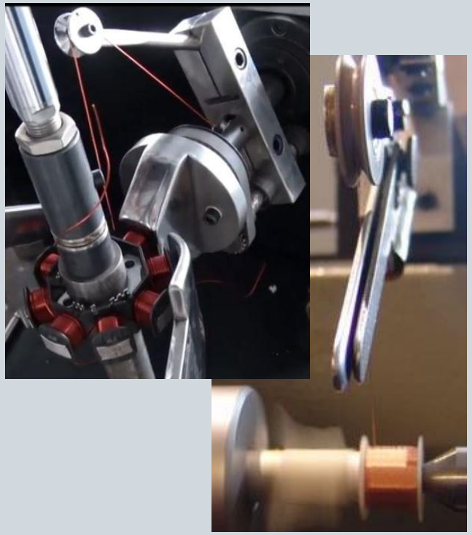
Needles

- Straight needles
- Curved needles

Graphics: Marsilli

Metal sheets

- Flyer winding blocks
- Sheet metal channels



Graphics: FAPS

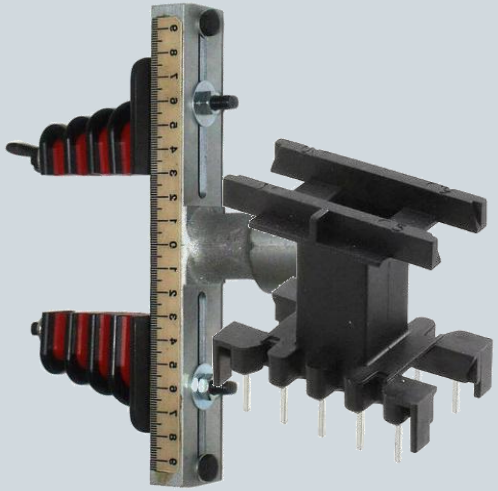
The bodies to be wound vary greatly depending on the product design.

Winding body

Graphic: PVS

Coil / stencil

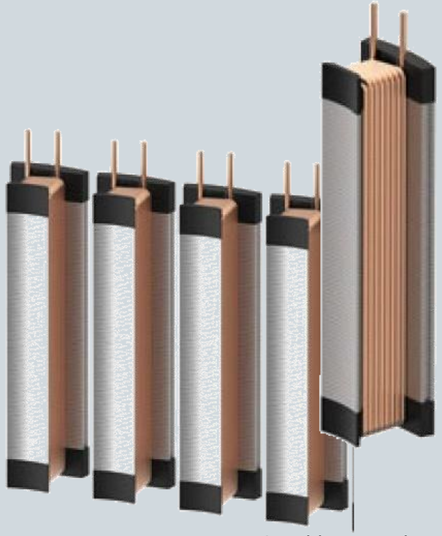
- Coil carrier
- Winding stencil



Graphic: RS, karl-wetzel

Stator tooth

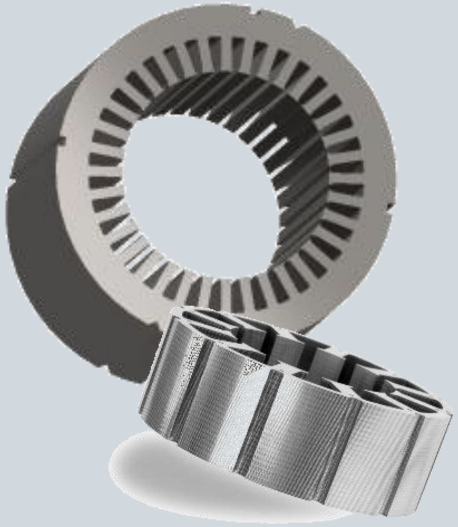
- Single-tooth
- Pole-chain



Graphic: Hagedorn

Lamellar stack

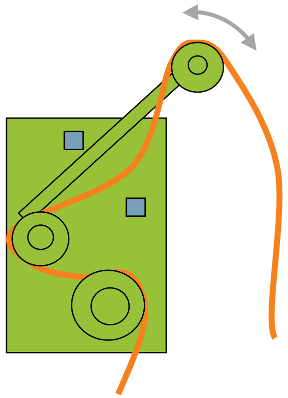
- Rotor lamellar stack
- Stator lamellar stack



Graphic: Vacuumschmelze

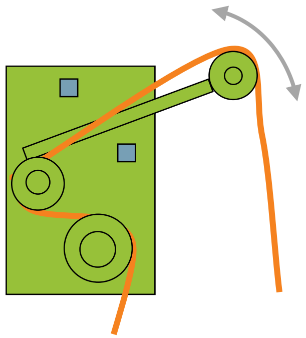
The wire tension is the most important influencing factor for forming the conductive material during the winding manufacturing process.

Wire tension too low



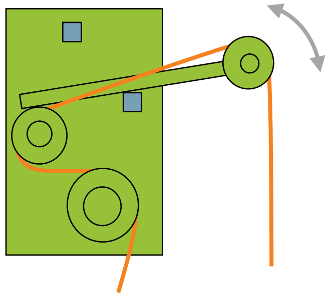
- Compensating element sits at upper end position
- Generates loose windings
- Max. fill factor cannot be achieved
- Low mechanical stability in the winding
- Weakened electromagnetic properties

Ideal wire tension



- Compensating element varies generously between both end positions
- Ensures packing density during wire laying
- Wire remains under tension
- Compensates for free wire lengths resulting from the wire inertia and asymmetry of the winding bodies

Wire tension too high



- Compensating element sticks on the lower position
- Wire experiences plastic deformation
- Electrical resistance rises resulting from cross section reduction
- Weakened electromagnetic properties
- Wasted material because of wire breakage

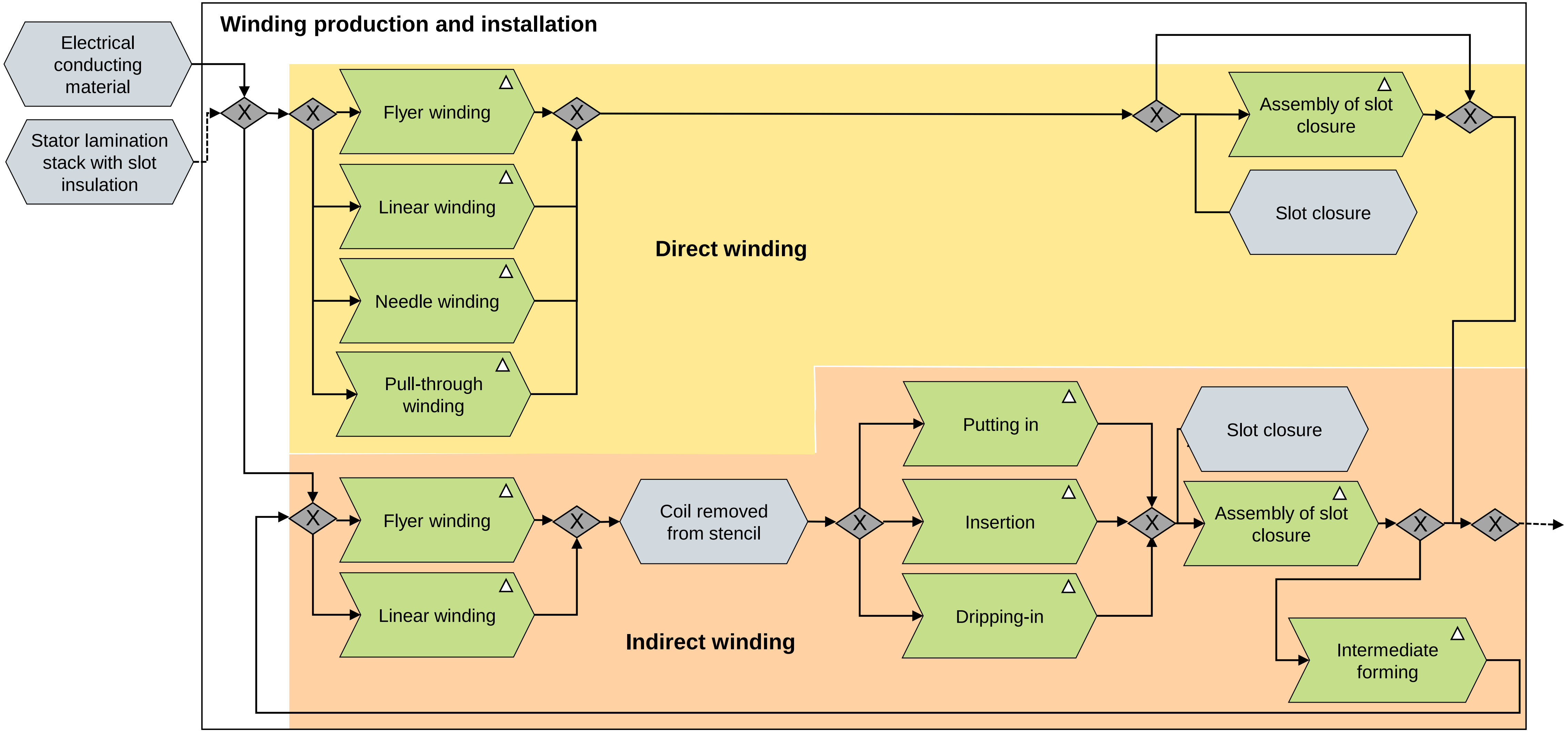
Lecture unit 6 provides an overview of the design and production steps for direct and indirect winding processes.

- Technical terminology
  - Definition of winding components and schematic diagrams
  - Significance of packing density and fill factor
  - Important characteristics of multi-phase alternating current windings

- **Winding processes and corresponding production system technology**
  - Components of a winding system
  - **Production diagrams for different winding processes**
  - Integration of the most common winding processes into the production chain for stators

Graphic: ZF Friedrichshafen AG

Different process variants are used for winding production and installation in the winding body.



There are basically six process variants available for the automated production of windings.

Production of windings

- Winding process
  - Linear winding
  - Flyer winding
  - Needle winding
  - Toroidal winding
- Winding production through assembly
  - Preformed coil/ bar winding
  - Drawing-in technique/ trickling

Linear winding

- Axially symmetrical coilformers
- Relays, chokes, transformers

Flyer winding

- Complex coilformers
- Rotor, SMD Coils, transformers

Needle winding

- Direct winding of multi-pole coilformers
- Stators and rotors

Toroidal winding

- Closed, ring-shaped coilformers
- Chokes, transformers

Preformed coil/ bar winding

See lecture unit F7

- Stators and rotors
- Traction drives, large drives, generators

Drawing-in technique/ trickling

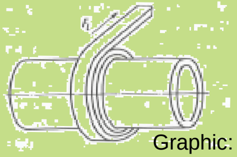
- Winding the coils onto the template
- Pulling in the coils
- Stators

Graphics: Kaschke, SEW, FAPS, Westdeutsche Dochtfabrik, ATOP

In linear winding, the wire is guided onto a rotating coil body by a linear travelling guide element.

Linear winding process

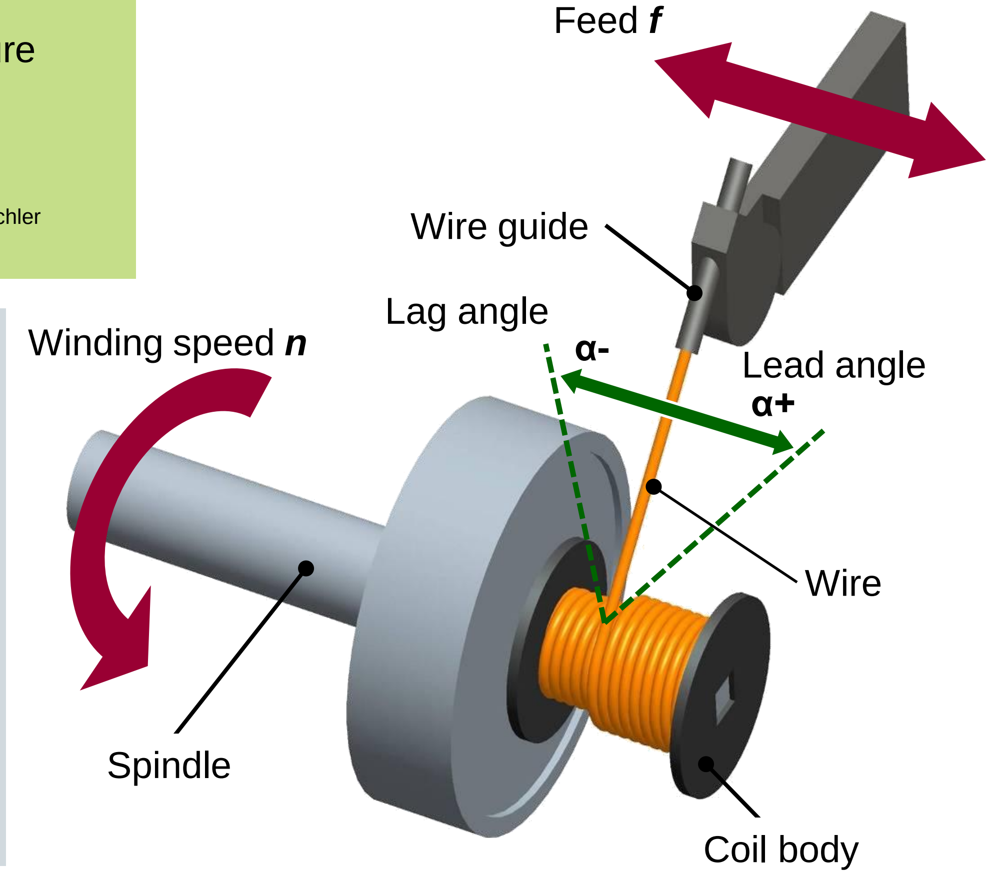
- Wires fed from a single wire guide
- Coil body rotates about the z-Axis ⇨ radial winding structure
- Linear motion of the wire guide along the z-Axis ⇨ axial winding structure
- Without feed motion, disc coils are produced:
- Optional: Use of guide plates



Graphic: Tzscheutschler

Video: Aumann

https://www.youtube.com/watch?v=cq5vRQEQMbw



The linear winding process represents the most productive winding process due to its many advantages.

Advantages of linear winding process:

- Highest rotational speed up to 30,000 RPM
- Wire thickness up to max. 5 mm diameter
- Applicable for shaped wire (flat wire)
- Very high fill factors achievable through orthocyclic windings
- Wire stress from machine feed largely avoidable
- Automatic wire positioning onto contacting sites possible
- Able to maintain lower resistance tolerances
- Multispindle arrangement possible

Disadvantages of linear winding process:

- Restricted to axially symmetric winding bodies
- Winding site must be accessible
- Only suitable for single-tooth (concentric) winding designs

Graphic: Kaschke

Areas of application:

Relays, chokes, transformers, RFID, concentric windings on stator-single teeth, coils for setting processes

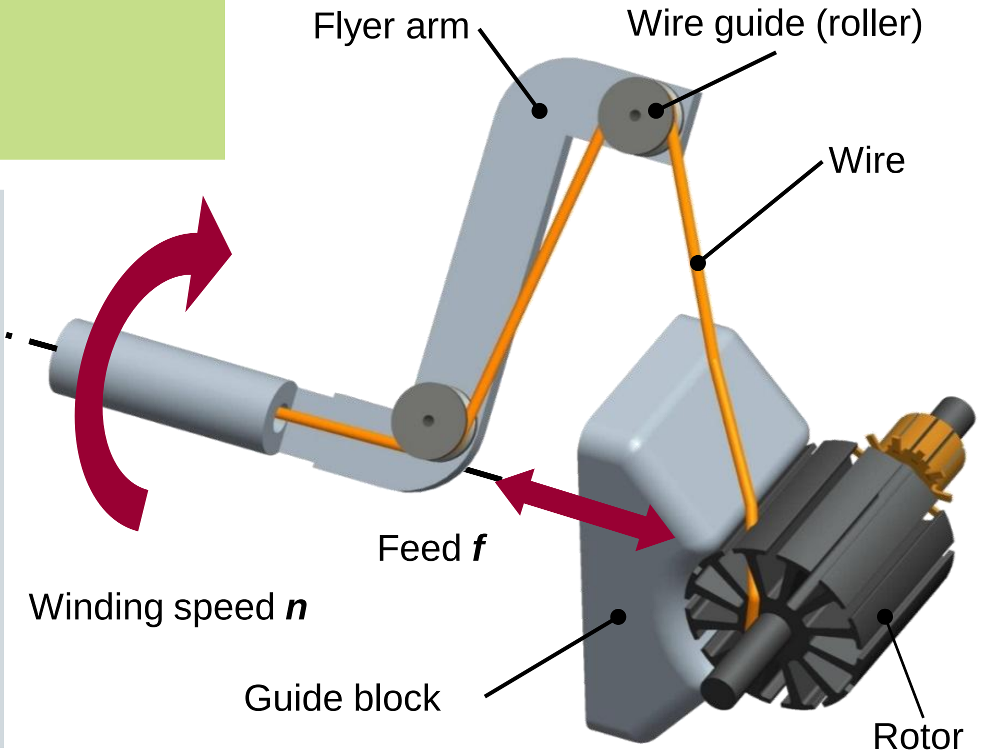
During flyer winding, the wire is guided from a rotating flyer to a fixed winding body.

Flyer winding process

- Wire guided over a flyer arm
- Flyer rotates about the z-axis
- Coil body fixed for one coil winding at a time, then rotated by one slot
- Winding wire slides over guide block into the winding chamber
- Optional relative motion in the z-axis

Video:
Nide

https://www.youtube.com/watch?v=Y_QEw1nUo68



The flyer winding process is especially suited for windings of thin wire with a high number of turns.

Advantages of flyer winding process:

- High flyer rotation up to 2,500 RPM
- Wire thickness up to 2.5 mm diameter with low speed
- Medium fill factor achievable
- Cost efficient process for a large number of turns
- Ideal application for small wire diameters
- Direct winding of complex coil bodies possible
- Automated interconnecting with a nozzle flyer possible
- Multispindle arrangement possible

Disadvantages of flyer winding process:

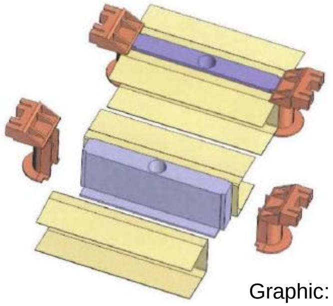
- Higher wire stress from torsion in flyer arm
- Operation with shaped wire not possible
- For direct winding on the coil bodies, only wild windings achievable

Areas of application:

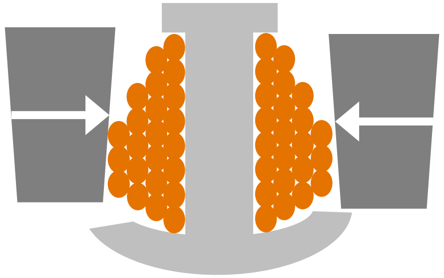
Transformers, SMD-coils, E-motor rotors, concentric windings on stator single-teeth, concentric and distributed windings on stators with outer slots, coils for setting processes

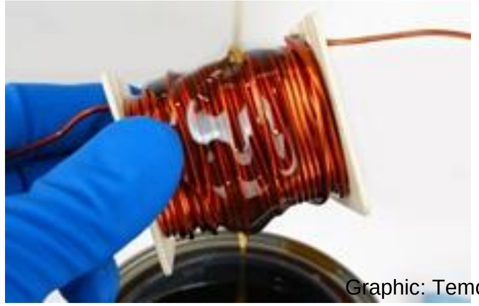
Stators with concentric windings can be cost effectively and efficiently manufactured with the linear or flyer winding process.

Material | Round or flat wire, polysterimide coated | Trapezoid shaped half-closed slots | Single pole segmented lamellar stack


Graphic: Jordan


Video: Marsilli




Graphic: Temco

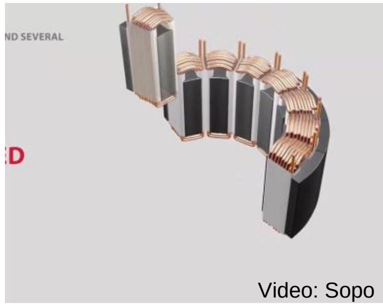
Face insulated parts installed

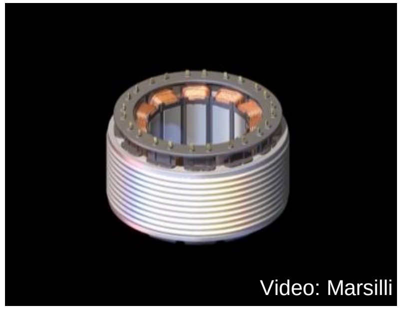
Slot base insulation inserted

Linear/flyer coil winding

Concentric winding calibrated

Single pole drip impregnation


Video: Sopo


Video: Marsilli


BMW

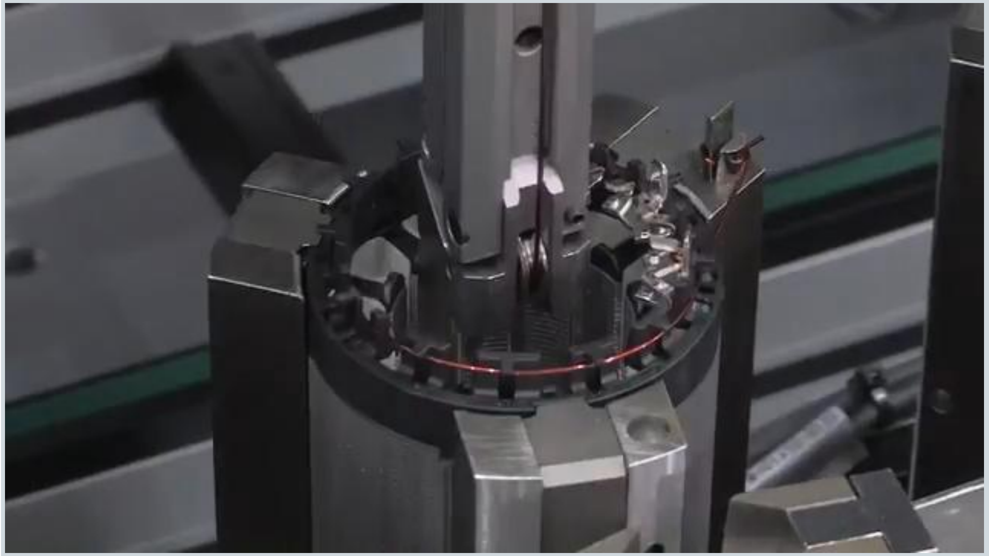
Wound single poles assembled

Coils connected by a circuit ring

Electric insulation testing

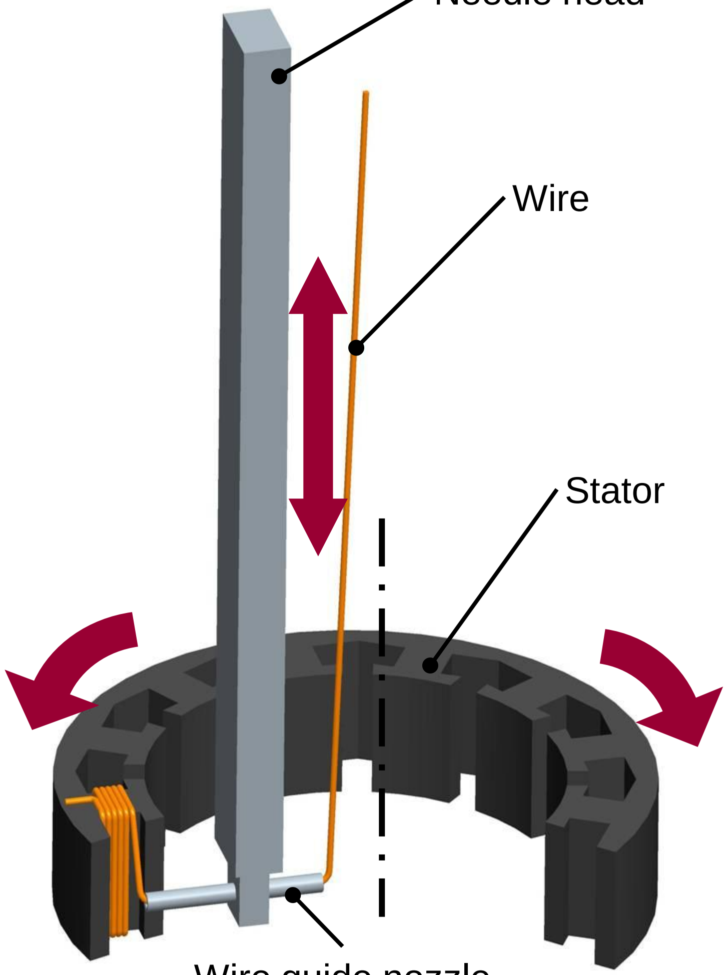
In needle winding, the wire is drawn out using a wire guide nozzle dipping into the slots of the winding body.

Needle winding process
■ Wire feed through wire guide nozzle (“needle”)
■ Stroke movement of the wire guide nozzle through the slot opening ⇨ Winding formation in the slot
■ Pivoting movement of the winding body by one pole pitch ⇨ Winding formation in the winding head
■ Optional pivoting of the needle winding head



Video:
aumann

https://www.youtube.com/watch?v=OrJZcJdwoPs



Due to the flexible wire insertion method, the needle winding process allows complete assembly of stators and rotors on one machine.

Advantages of the needle winding process:

- Large wire thickness, up to 2.5 mm diameter
- Linear winding possible (copper fill factor under 40%)
- Direct winding of multipole coil bodies possible
- Automatic contacting of the phases
- Reduction of contacting sites possible (“through windings”)
- No follow-on process for forming the winding head required
- Stator segmenting can be bypassed
- Complex wire guiding possible

Disadvantages of the needle winding process :

- Various high bending stresses in the wire
- Operation with shaped wire not possible
- Space required for the needle tool reduces the achievable fill factor
- Low speed: up to 600 strokes/min

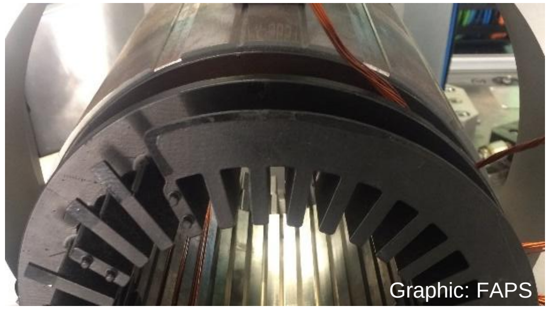
Graphic: SEW

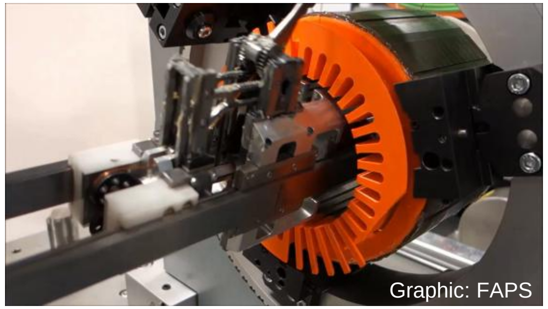
Areas of application:

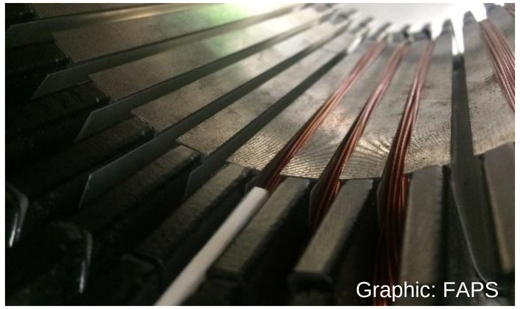
Concentric and distributed windings in stators and rotors with both outer and inner slots

Needle winding technology enables a fully automated manufacturing chain, but is only possible with an adapted stator design.

Material | Polysterimide coated round wire | Trapezoid shaped half-closed slot | Unsegmented lamellar stack


Graphic: FAPS


Graphic: FAPS

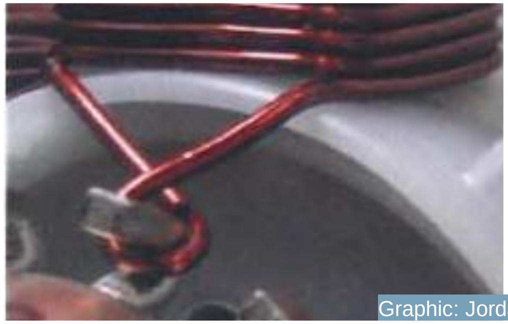

Graphic: FAPS

Attach face insulation disc with glue

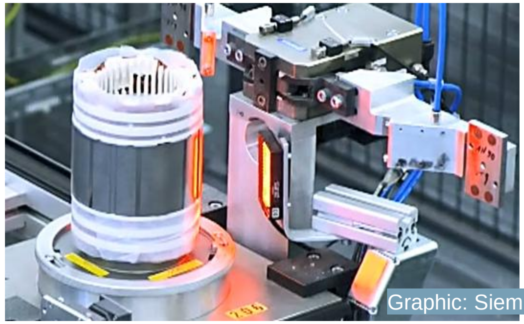
Insert slot base insulation

Needle winding

Insert phase insulation


Graphic: Jordan


Graphic: FAPS


Graphic: Siemens

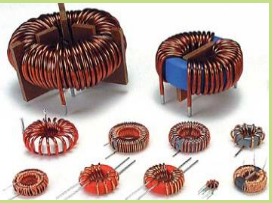
Windings contacted through welding

Electric insulation testing

Stator impregnation through current-UV

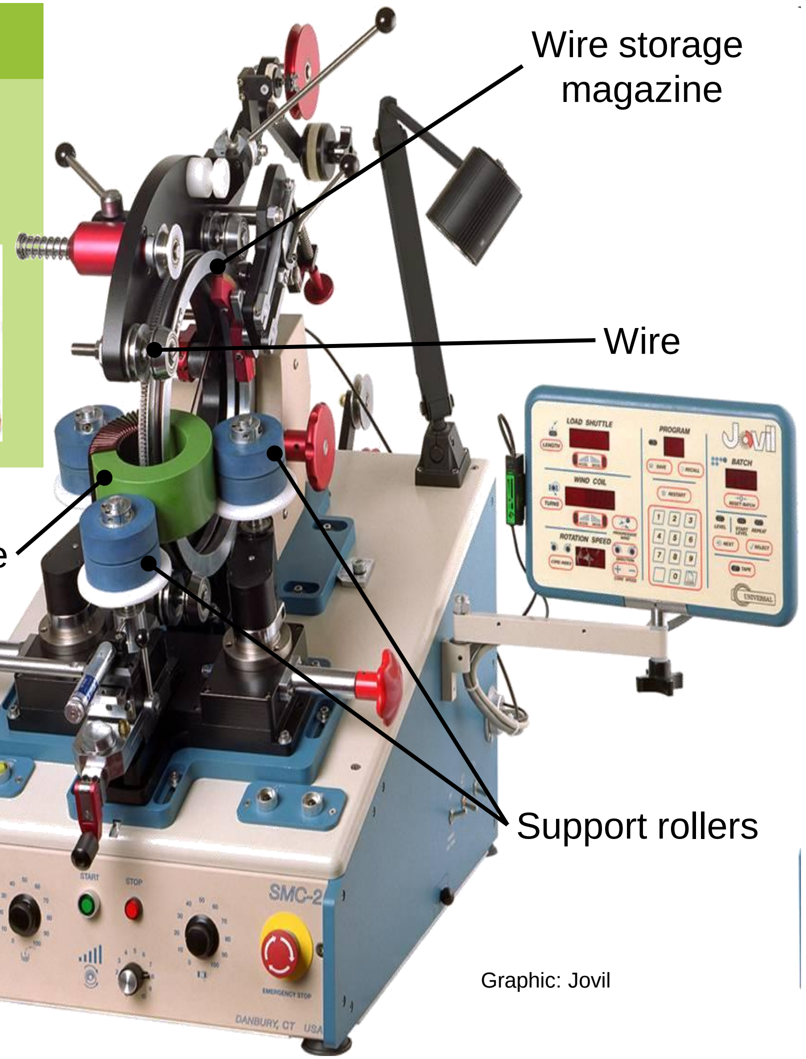
Using a round wire storage magazine, toroidal coil bodies can also be wound.

Toroidal core winding process
■ Feed movement of the toroidal core by rotating three-point support
■ Wire placed onto toroidal core through by rotating the wire storage magazine
■ Direct winding of closed, toroid shaped coil bodies
■ High degree of manual labor required
■ Closed, toroid shaped coil body
■ Chokes, transformers



Video:
UE Technologies

https://www.youtube.com/watch?v=nH8A4ohj-QQ

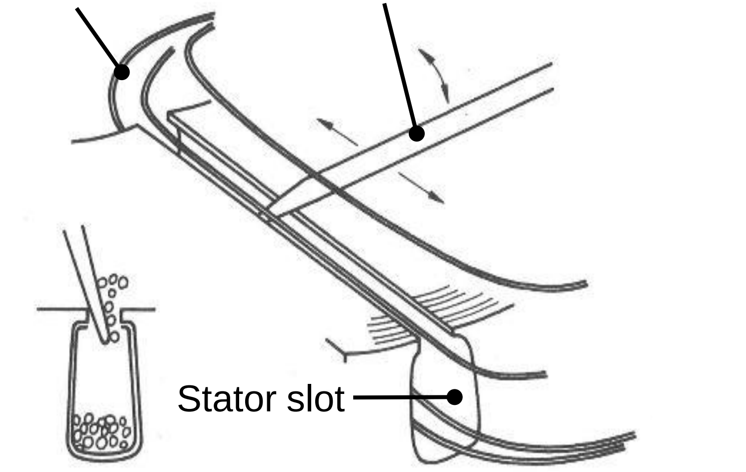


Graphic: Jovil

During dripping-in, the wires of a prefabricated coil are inserted manually into the slot one after the other.

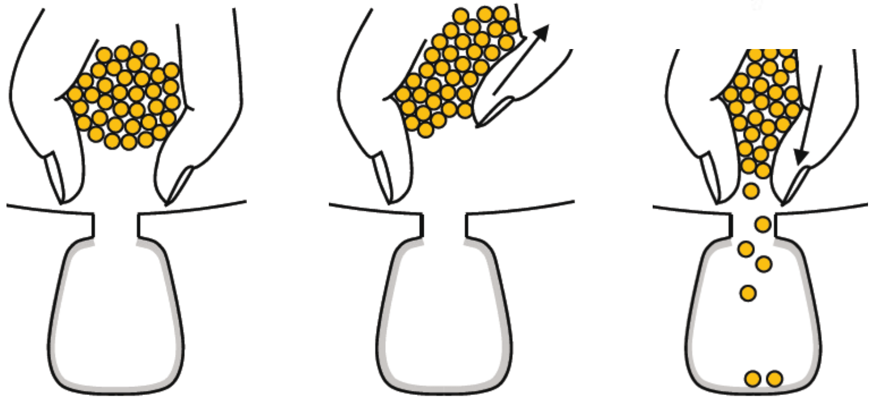
Coil assembly through drip-in process

- Coils produced in separate winding process (Linear, Flyer)
  ⇨ indirect winding process
- Assembly via axial insertion of the coils into the interior of the winding body
- Coil sides are divided into individual wires in the radial direction
- Individual wires of the spool side are individually dripped through the slot opening into the winding slot
- Additional tools such as drip needles or wooden wedges may be used



Video: Hurricane

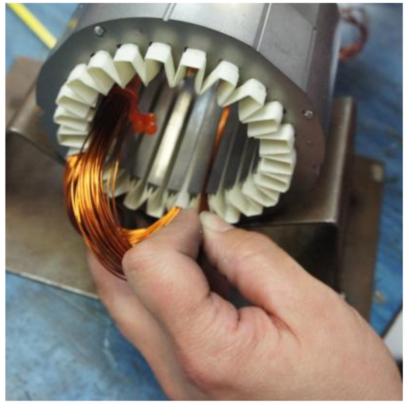
https://www.youtube.com/watch?v=bCwu5KPVv54



Placement of the winding over the slot

Reduction of the winding cross section

Dripping-in the individual wires



The assembly of large-scale electric machines is characterized by manual operations.

Process

Manual stator assembly

Video:
Science Channel

youtube.com/watch?v=eauUQ8N63kE

The dripping-in process is only suitable for very small quantities due to the high manual and time-consuming production work involved.

Advantages of the dripping-in process:

- High fill factor achievable
- High degree of freedom in the winding scheme design
- Low investment cost through lack of machines


Graphic: SEW

Disadvantages of the dripping-in process:

- Long process time through manual assembly
- High operating cost for skilled labor
- Reduced repeatability and process reliability
- Quality heavily dependent on employee skill

Areas of application:

Small series production, specialty motors, prototype production, repair

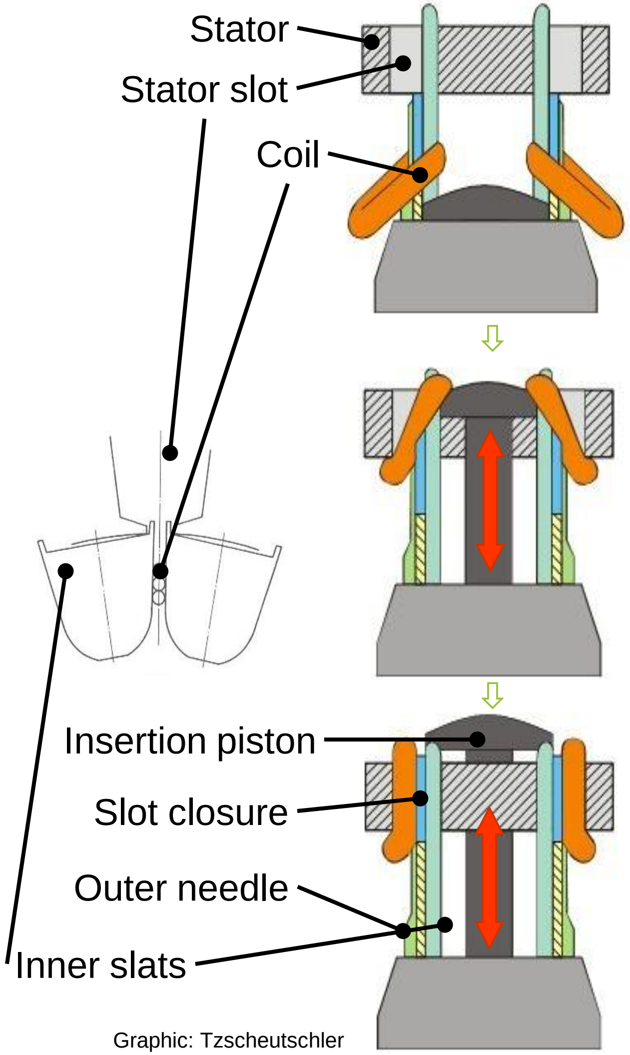
In the Insertion process, the wires of a pre-wound coil are drawn into the slots by means of a insertion piston.

Insertion process
Coil production in separate process (Linear, Flyer) (or direct on the tool needles)
Placement of coils between the insertion tool needles
Placement of the stator onto a setting tool
Axial stroke of insertion piston exerts pressure onto the coils
Wires are pulled into the stator slots between the insertion tool inner needles
Slot closures and phase separators are inserted after the stroke movement

Video:
NIDE Group

https://www.youtube.com/watch?v=nCF1hM9mmmU

Graphic: Otto Rist Gmbh



Graphic: Tzscheutschler

The stator serves to create the rotating magnetic field.

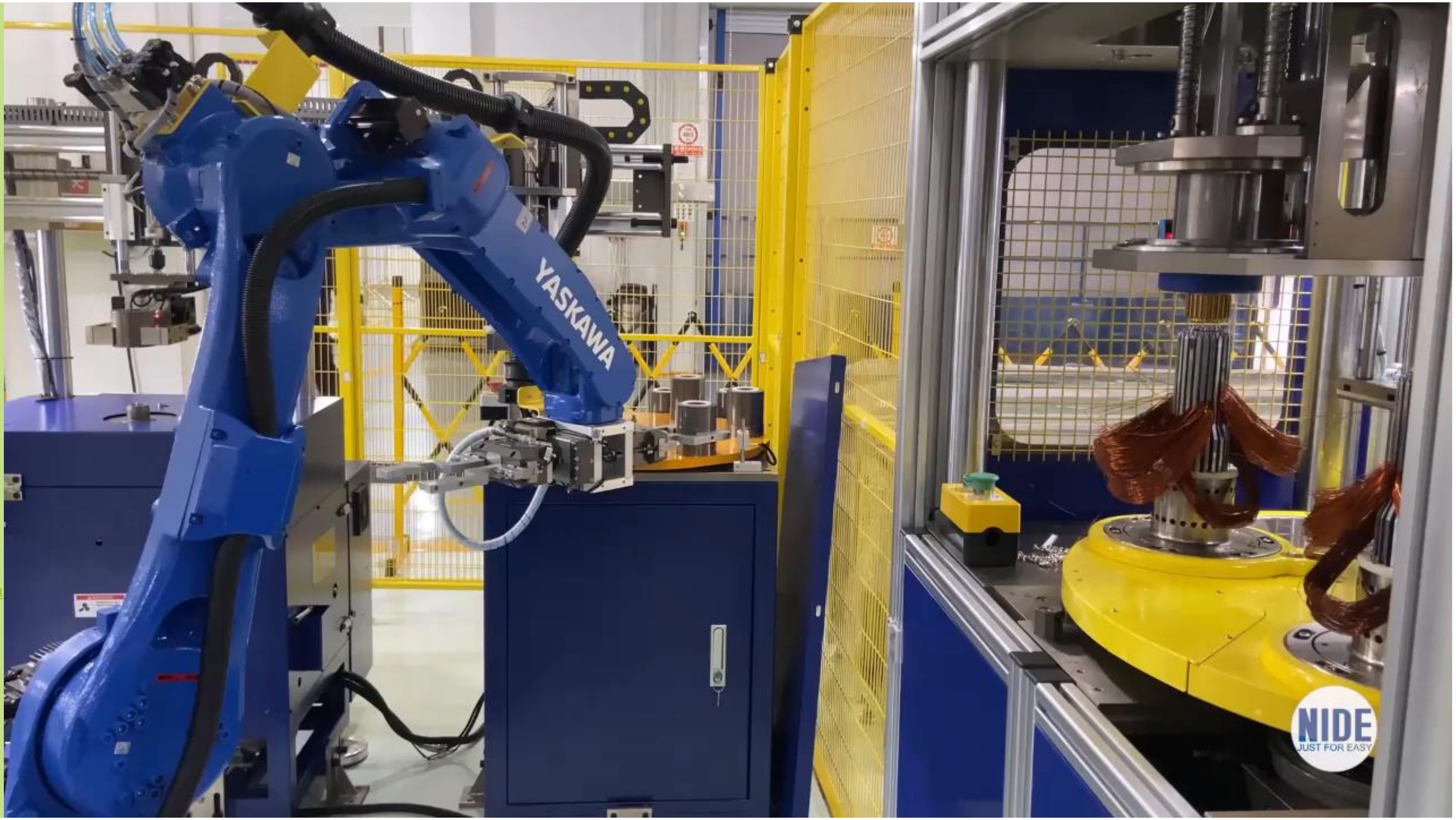
Slide from Lec. 1

Process

Automatic stator production line for electric induction motor

Video:
Nide Group

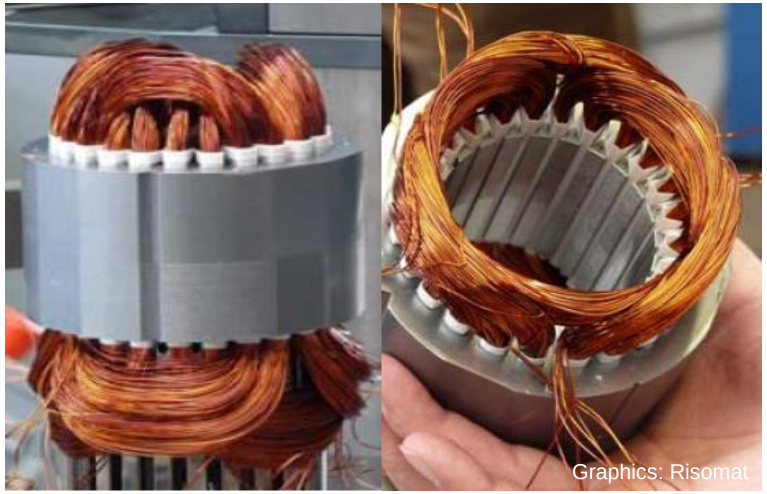
https://www.youtube.com/watch?v=Jikiy4KLO0k



The insertion process represents the most productive setting process for the largely automated production of complex winding schemes.

Advantages of the insertion process:

- High fill factor achievable (over 50% copper fill factor)
- Complex winding schemes with thin round wire possible (distributed windings, double-layer windings, many parallel wires, multi-level windings)
- Equal coil lengths achievable
- Stacked sheets can be interlocked afterwards



Graphics: Risomat

Disadvantages of the insertion process:

- High investment costs for tools and machines
- No positional accuracy in winding production
- Wire stress during insertion process
- Operation with shaped wire not possible
- Manual and machine reworking necessary
  (phase insulation, winding head lacing and forming)

Areas of application:

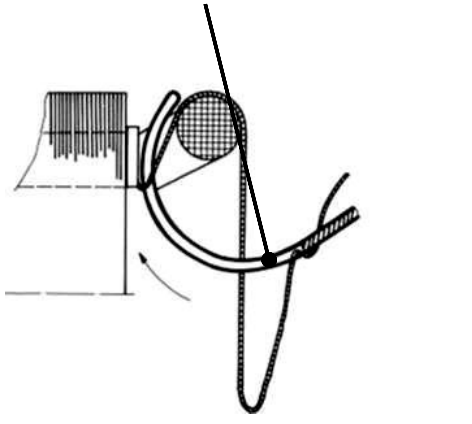
Middle and large series production, distributed windings in stators with inner slots, double-layer windings

By lacing the preformed winding heads with twine, coil transitions, switching connections and phase insulation parts are fixed.

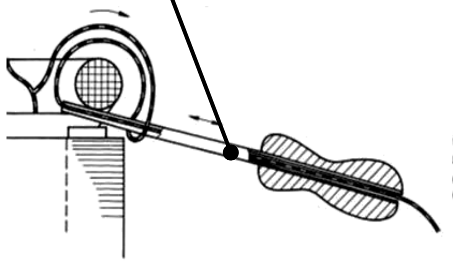
FAPS

Manual lacing

Steel wire lacing needle



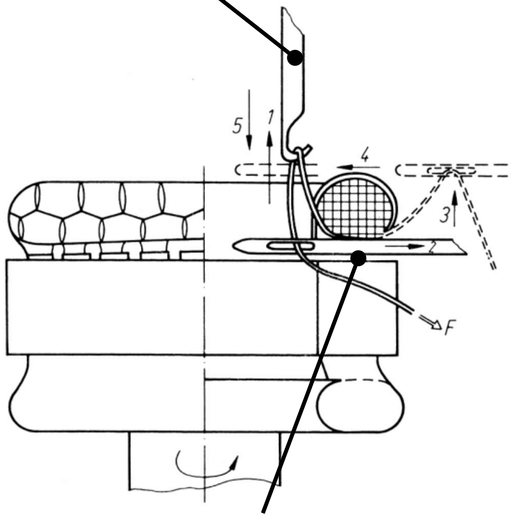
Pipe lacing needle



Graphic: Tzscheutschler

Automated lacing

Loop lifter



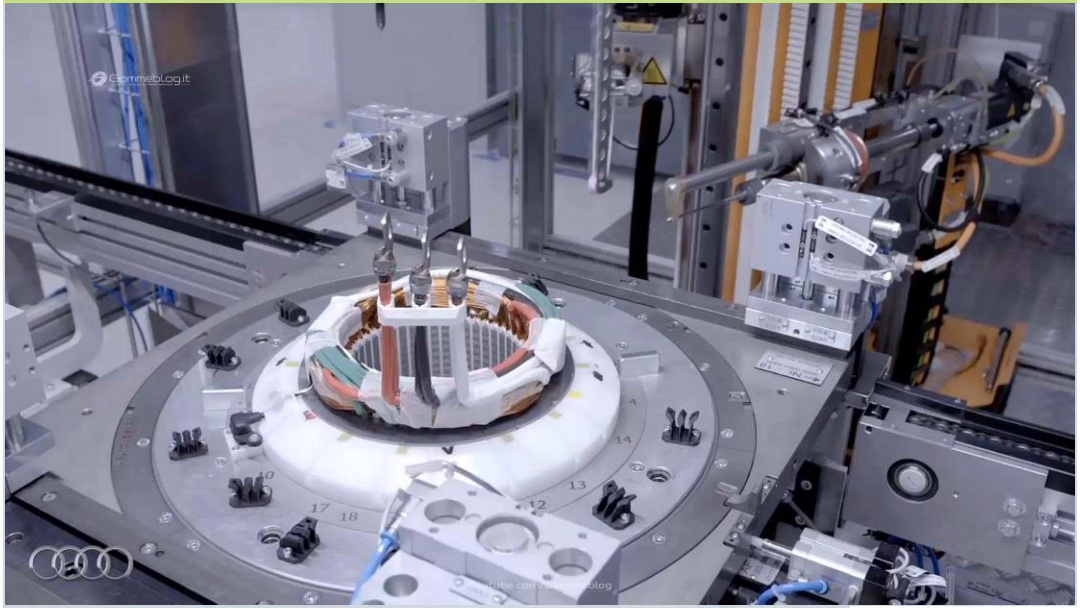
Lacing needle

1 – 5: motion sequence

Graphic: Tzscheutschler

Process

Automated lacing on a round wire winding head following an insertion process



Only required for dripped-in or insertion windings

Video:
Audi und Renault

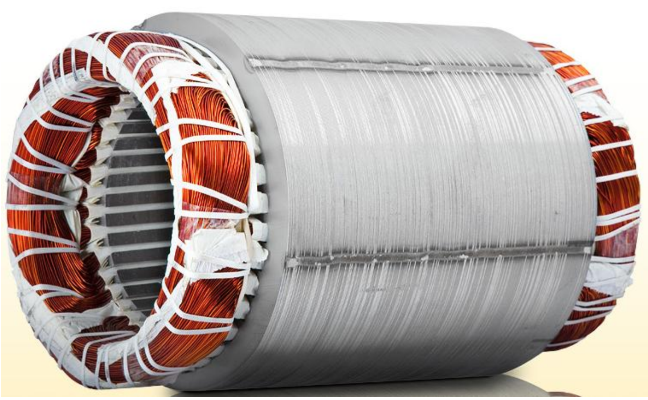
youtube.com/watch?v=8bbjb7_jX6c

FAPS

The shape and especially the height of the winding head are important features of electric drives.

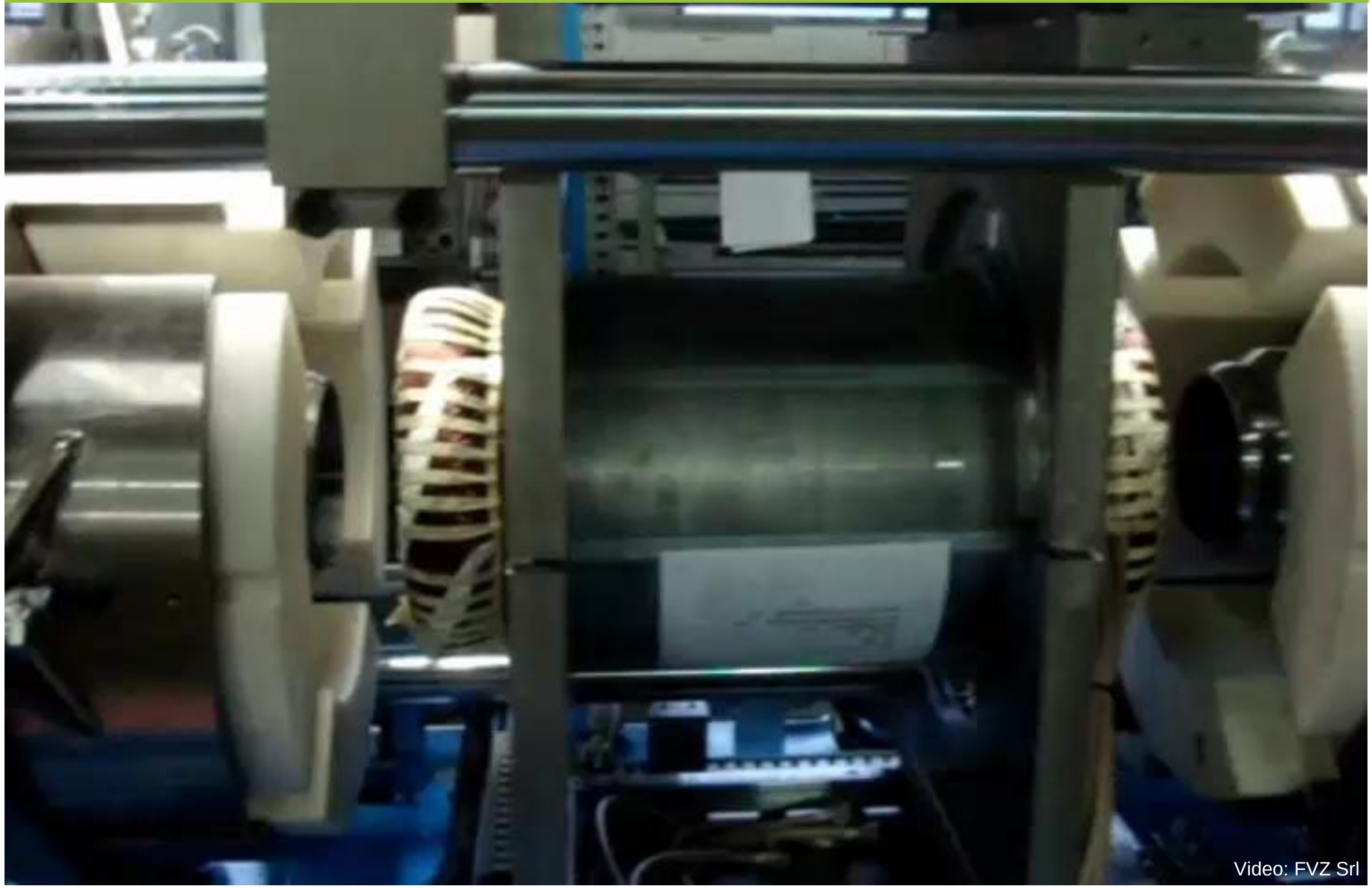
Pressing the winding head

- Typically used with round wire windings produced with indirect winding processes (insertion process, drip-in process)
- Creates a defined winding head form
- Reduces the space requirement of the winding head



Graphic: Renke & Müller GmbH

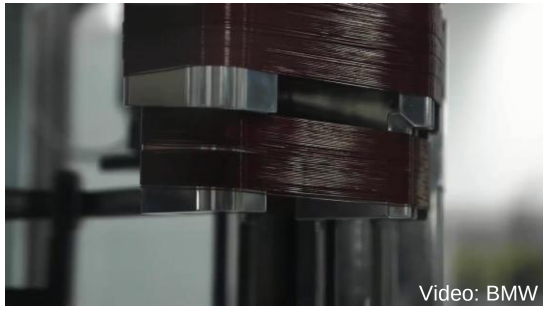
Process

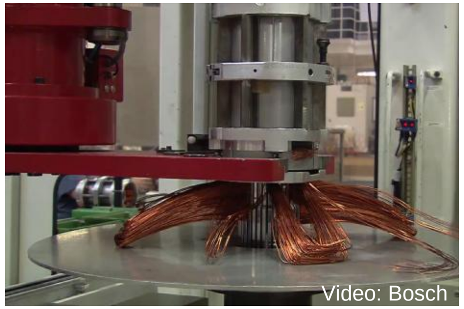


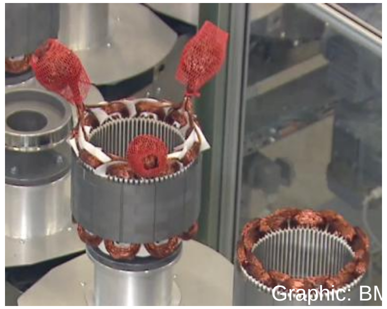
Video: FVZ Srl

Insertion technology is the most economical manufacturing method for distributed windings, but it reaches its limits at the automation level.

Material | Polysterimide coated round wire | Trapezoid shaped half-closed slot | Unsegmented lamellar stack

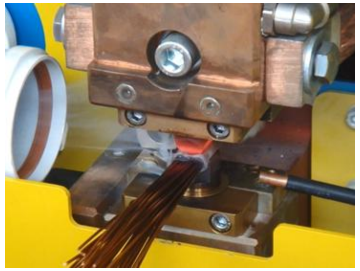
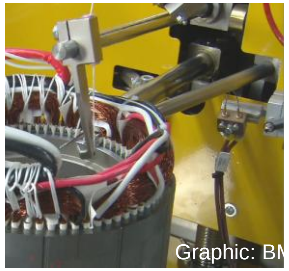


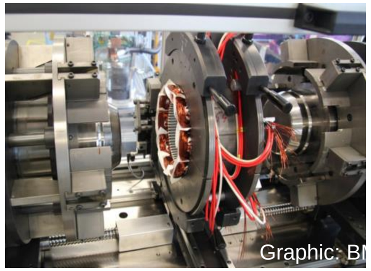


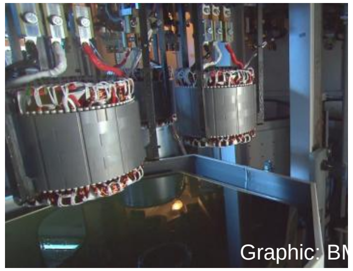


- Linear coil winding
- Insert slot base insulation
- Mount coils and slot closures
- Interim pressing winding head
- Attach phase insulation (M)

(M) Manual step & (P) Partially automated step





- Attach insulating tubes(M)
- Hot crimping phase contacts(P)
- Lace winding head
- End pressing winding head
- Electric insulation testing(P)
- Current-heat impregnation

Lecture unit 6 provides an overview of the design and production steps for direct and indirect winding processes.

- Technical terminology
  - Definition of winding components and schematic diagrams
  - Significance of packing density and fill factor
  - Important characteristics of multi-phase alternating current windings

- **Winding processes and corresponding production system technology**
  - Components of a winding system
  - Production diagrams for different winding processes
  - **Integration of the most common winding processes into the production chain for stators**

Graphic: ZF Friedrichshafen AG

The design of the electric motor determines the winding method to be used and the complexity of the winding process.

FAPS

Example: concentric windings

Winding bodies

Internal rotor

Rotor

Graphic:: Aumann

Stator

Graphic: Aumann

External rotor

Stator

Graphic: Aumann

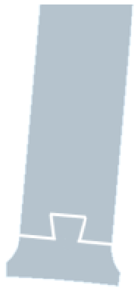
Internal and external rotor

Stator

Graphic: CPM

Possible methods for segmenting the lamellar stack

Single pole shoe



Pole shoe shell

Graphic: Jordan

Single segment

Graphic: Jordan

Pole chain

Graphic: Jordan

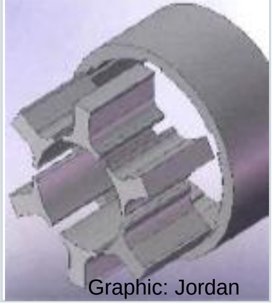
Unsegmented

Graphic: Jordan

Pole star

Graphic: Jordan

Single tooth

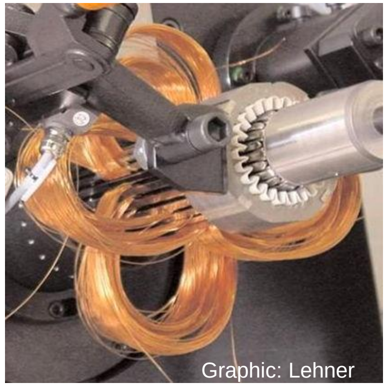


Graphic: Jordan

To produce <u>distributed</u> windings, three different processes are possible, each with specific advantages and disadvantages.

FAPS

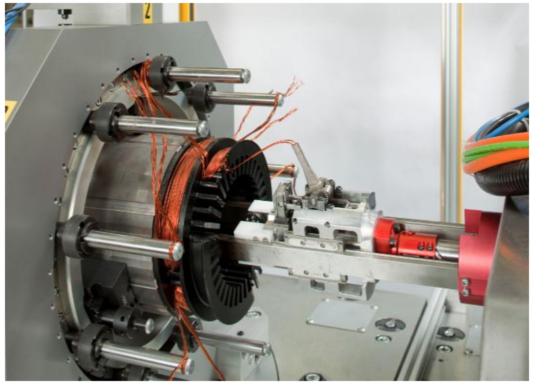
	Indirect winding process with corresponding assembly process		Direct winding process
	Insertion technique	**Preformed coil technique**	**Needle winding technique**



Graphic: Lehner

Graphic: Micamation

See lecture unit F6



	Insertion technique	Preformed coil technique	Needle winding technique
Advantages	■ High degree of freedom in winding design ■ Low cycle time achievable	■ High degree of freedom in winding design ■ High slot fill factor possible ■ Defined wire position in the slot	■ Defined wire placement ■ Complete automation possible ■ Can produce many variants in one device
Dis-advantages	■ Wild windings ■ Not completely automated ■ Limited slot fill factor ■ High tool cost	■ Limited slot shape designs ■ Square wire cross section ■ Many contact points	■ Long cycle time ■ Overhanging winding head ■ Low degree of freedom in winding design

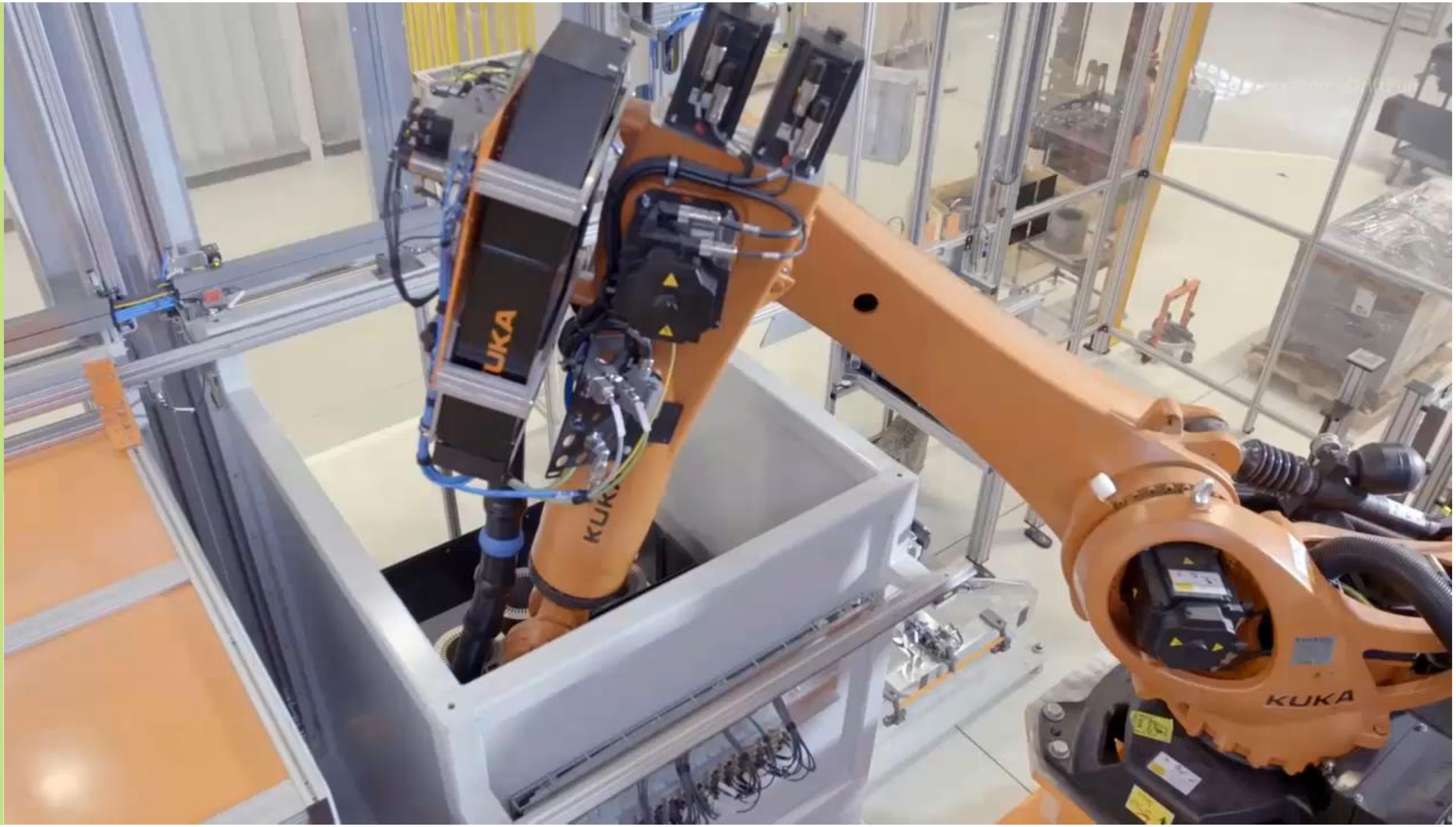
The electric drives for Audi e-tron models are manufactured with a high degree of automation.

Process

Automated production of stators for automotive traction drives

Video: Audi

https://www.youtube.com/watch?v=cd5iuv4ENOE



The contents of the lecture unit can be further deepened with the following specialized literature:

Textbooks

- Much W.; et al (1986): „Wicklungen und Montage rotierender elektrischer Maschinen“, VEB Verlag Technik, Berlin, ISBN 3-341-00020-8.
- Bălă C.; et al. (1976): „Handbuch der Wickeltechnik elektrischer Maschinen“, VEB Verlag Technik, Berlin.
- Tzscheutschler, R., H. Olbrich und W. Jordan. Technologie des Elektromaschinenbaus. Berlin: Verl. Technik, 1990. ISBN 3-341-00851-9
- Müller, Germar; Vogt, Karl; Ponick, Bernd (2008): Berechnung elektrischer Maschinen. 6., völlig neu bearb. Aufl., 1. Nachdr. Weinheim: Wiley-VCH Verlag GmbH & Co. KGaA, Weinheim (Wiley Interscience Online Books, 2).
- Sequenz, Heinrich (Hg.) (1973): Herstellung der Wicklungen elektrischer Maschinen. Unter Mitarbeit von M. Brüderlink, E. Feiten, O. Haus, R. Knobloch, D. Lambrecht, F. Maier et al. Wien: Springer. Online verfügbar unter https://www.springer.com/de/book/9783709143827.
- Heiles, Franz (1936): Wicklungen Elektrischer Maschinen und Ihre Herstellung. Berlin, Heidelberg, s.l.: Springer Berlin Heidelberg.
- Richter, Rudolf (1949): Kurzes Lehrbuch der Elektrischen Maschinen. Wirkungsweise · Berechnung · Messung. Berlin, Heidelberg: Springer Berlin Heidelberg.
- Kampker, Achim; Kreisköther, Kai D.; Kleine Büning, Max; Treichel, Patrick (2018): Herausforderung Hairpintechnologie Technologieschub für OEMs und Anlagenbauer. In: ATZ Elektron 13 (5), S. 62–67. DOI: 10.1007/s35658-018-0061-6.

The contents of the lecture unit can be further deepened with the following specialized literature:

Dissertations

- Wenger, U. *Prozessoptimierung in der Wickeltechnik durch innovative maschinenbauliche und reglungstechnische Ansätze.* Dissertation. Bamberg: Meisenbach, 2004. Fertigungstechnik Erlangen. 147. ISBN 3-87525-203-9
- Dobroschke, A. *Flexible Automatisierungslösungen für die Fertigung wickeltechnischer Produkte.* Dissertation. Bamberg: Meisenbach, 2011. Fertigungstechnik Erlangen. 219. ISBN 3-87525-317-5
- Kühl, A. *Flexible Automatisierung der Statorenmontage mit Hilfe einer universellen ambidexteren Kinematik.* Dissertation. Bamberg: Meisenbach, 2014. Fertigungstechnik Erlangen. 251. ISBN 978-3-87525-367-2

Other Publications

- Bickel, Benjamin; Hübner, Markus; Franke, Jörg (2014): Analyse des Optimierungspotenzials zur Erhöhung des Kupferfüllfaktors in elektrischen Maschinen. In: ant Journal 53 (2), S. 16–21.
- Bickel, Benjamin; Mahr, Alexander; Meixner, Sebastian; Bäumler, Markus; Franke, Jörg (2016): Manufacturing Techniques for Improved Electric Traction Drives. In: Proceeding of the 26th International Conference on Flexible Automation and Intelligent Manufacturing (FAIM), Seoul, 27.-30.06.2016. Korea Science and Technologie Center, S. 520–527.
- Mahr, Alexander; Mayr, Andreas; Jung, Timo; Franke, Jörg (2019): Robot-assisted concept for assembling form coils in laminated stator cores of large electric motors. In: Elsevier (Hg.): Procedia Manufacturing Volume 38. 29th International Conference on Flexible Automation and Intelligent Manufacturing. FAIM 2019, Bd. 38.

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THANK YOU